

Mechanical Technology

Welding

Tools and Equipment

By the end of the topic learners will be able to:-
use measuring equipment and explain why it is so important that these tools are taken care of and properly stored away.

WHAT IS A TOOL AND WHY DO WE HAVE IT?



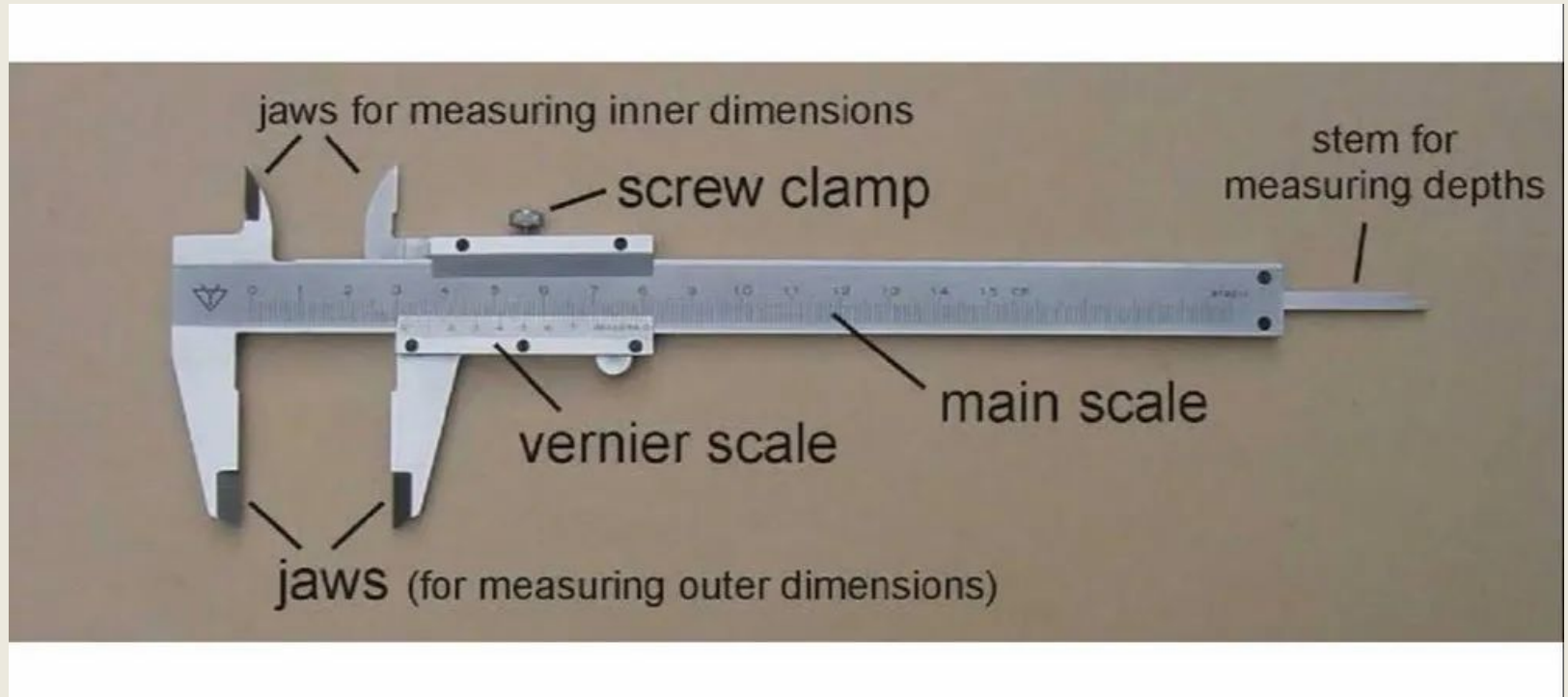
A tool is essentially an extension of the user's hand. That is why we say that your hands are your most important tools, that you cannot replace, so look after it!

Remember, always use the tool for its intended purpose. Use the tool for what it was made to do and do not abuse the tool.

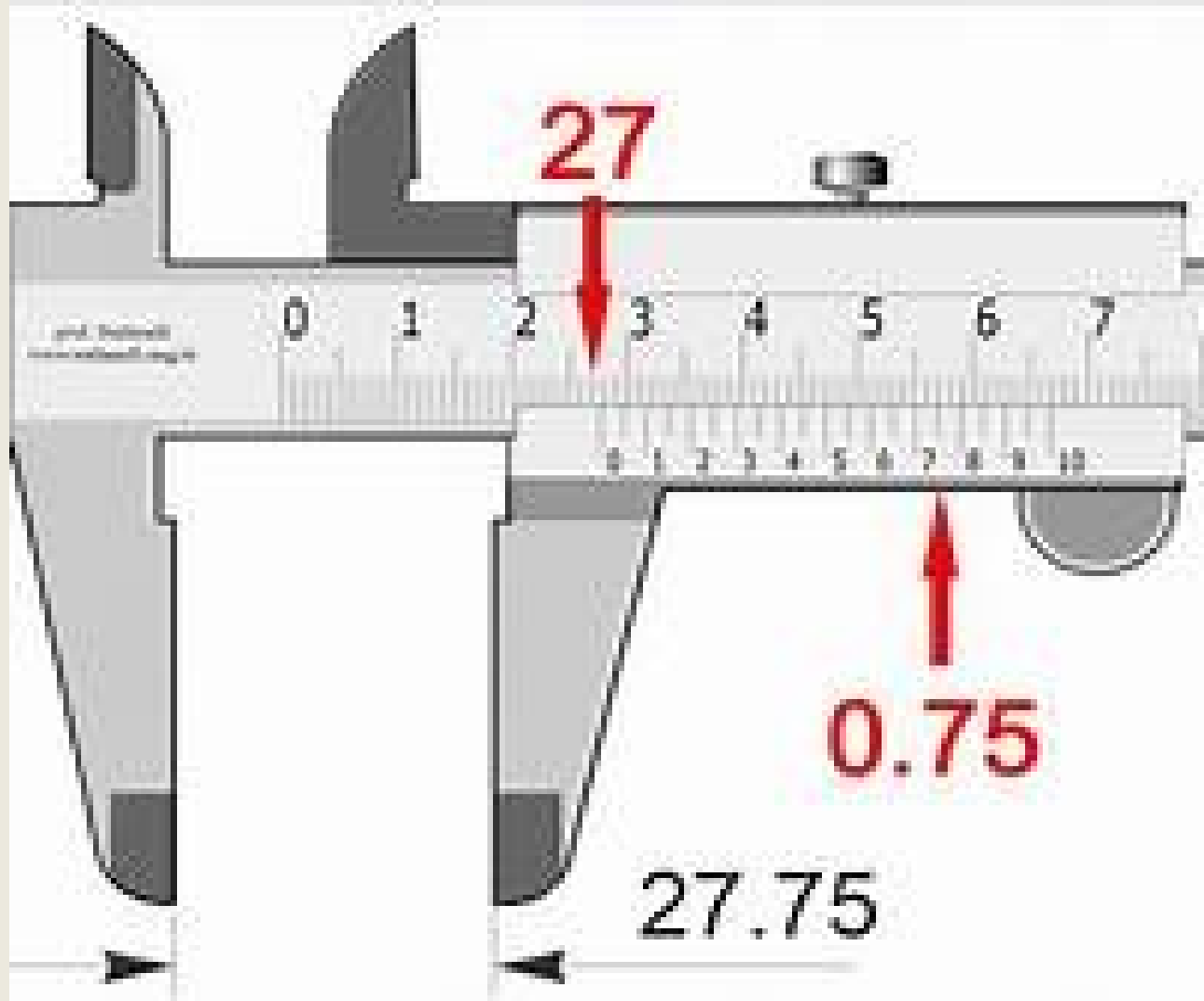
MEASURING



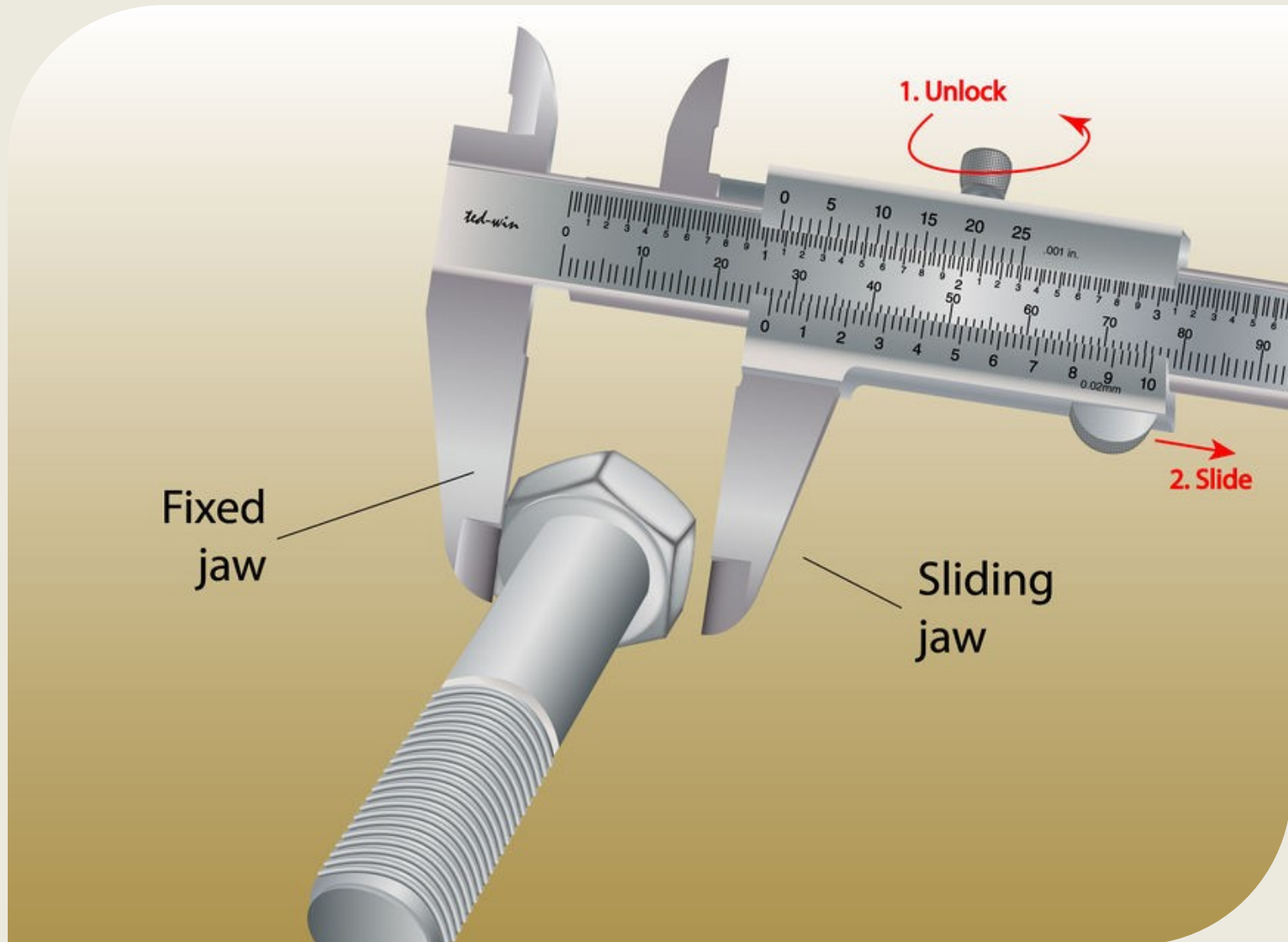
MEASURING



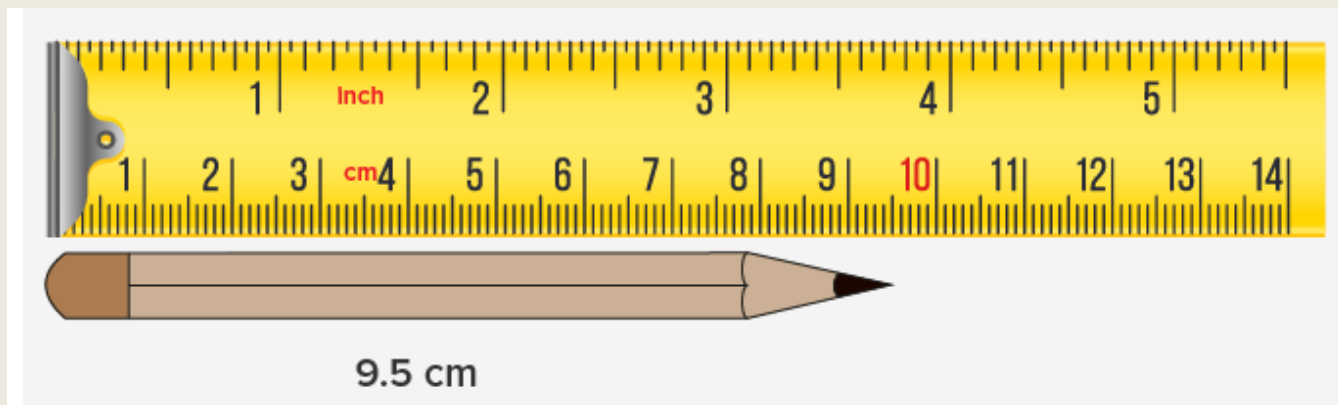
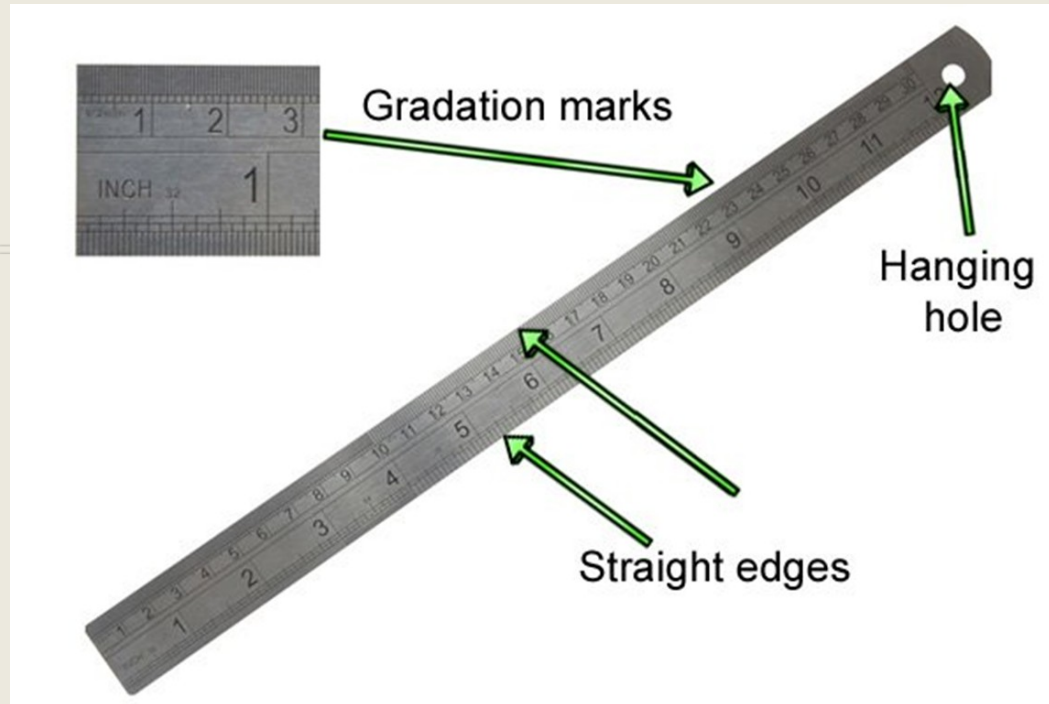
HOW TO USE A VERNIER CALIPER



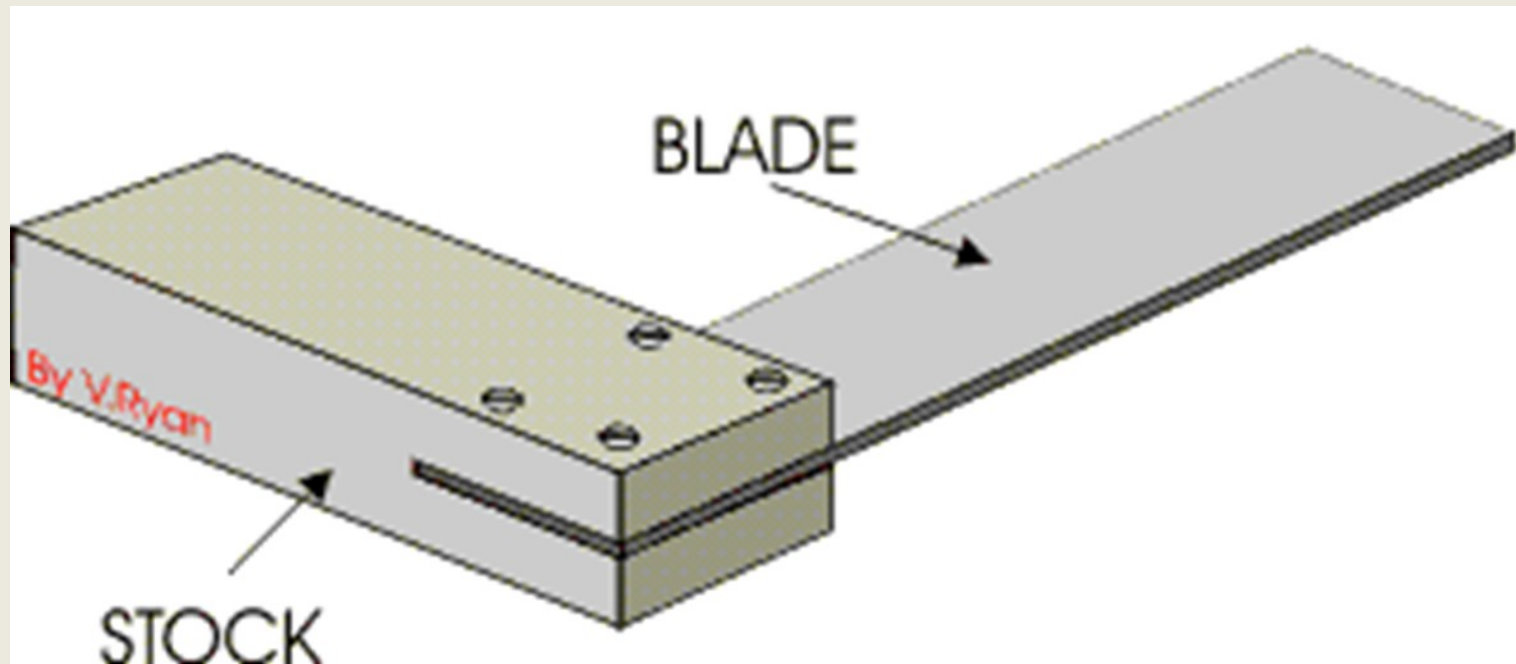
MEASURING



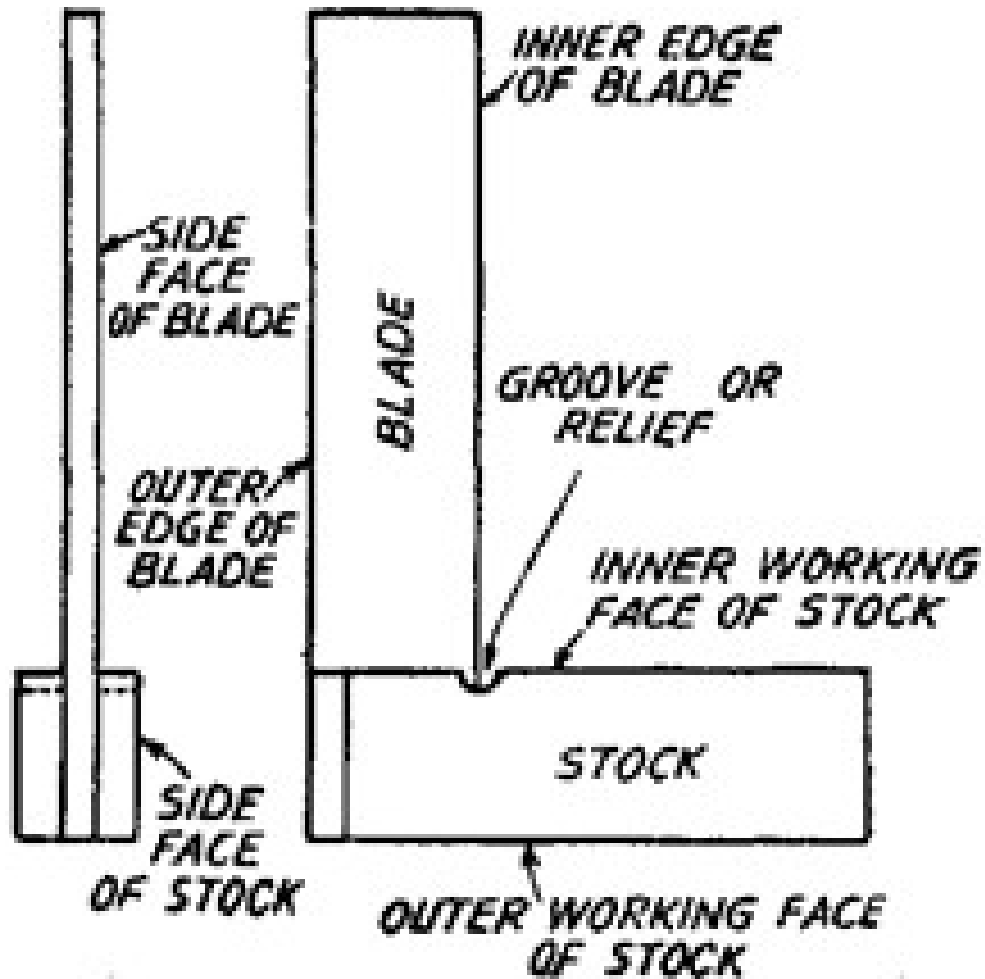
THE STEEL RULE



SET SQUARE

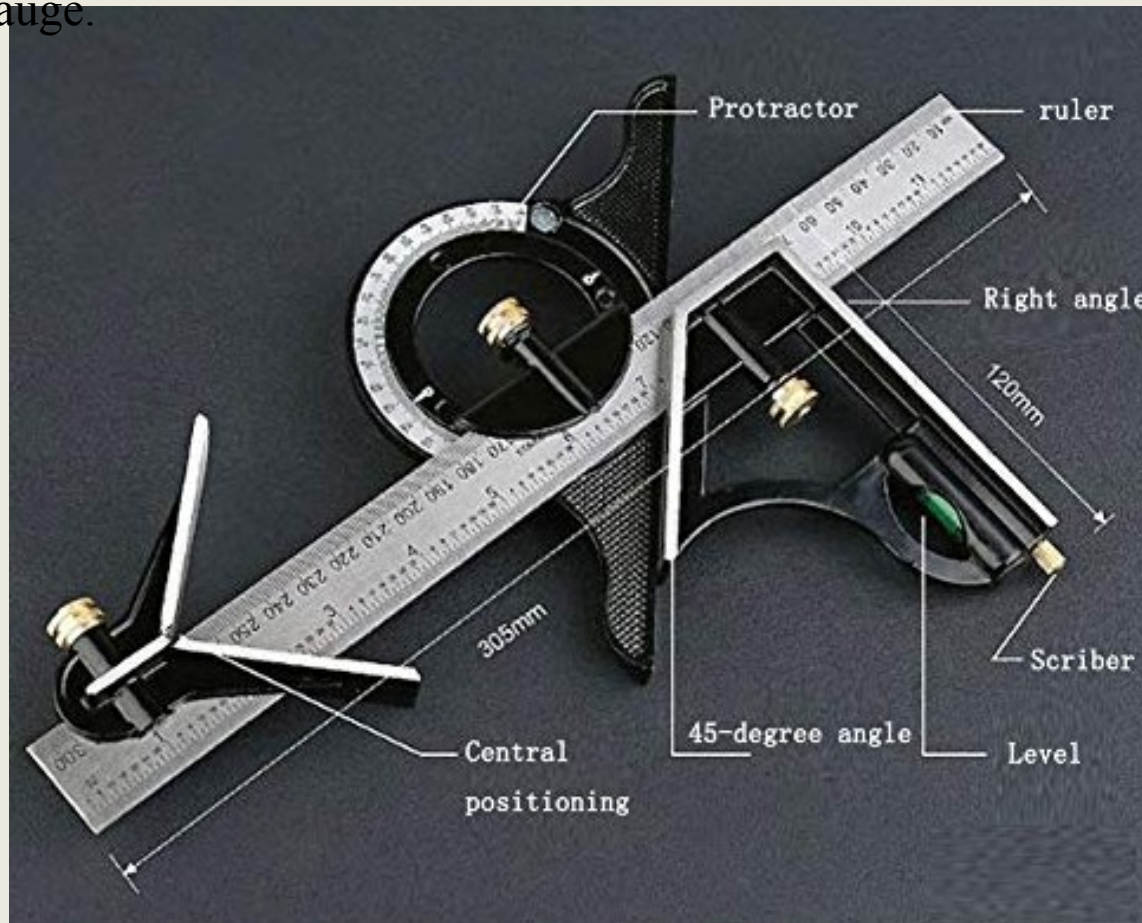


SET SQUARE



Multi-function combination set square

Multi-function 300mm/12" combination set which Can be used as a square, mitre, depth and height gauge.



DRILLING

The educator must explain and demonstrate the 5 point procedure of how a hole should be drilled into material.

- Measure and mark the spot
- Centre-punch the spot
- Drill a pilot hole
- Drill the hole that is required (finish hole)
- Chamfer the holes both sides to remove all sharp edges

Drilling Safely

- 1) Drill a pilot hole. In many cases, you'll get better results if you start with a drill bit a little smaller than the final hole size. Drill a shallow "pilot hole," then switch to the larger bit to finish the job. The pilot hole will help prevent your drill bit from slipping, and reduces the chance of splitting wood or other damage



2) Drill with steady pressure. Hold the drill steady and push it into the material you're drilling. If it takes more than light force to drill the hole, you're probably using the wrong bit



3) Adjust the clutch. Each drill has a twistable collar to adjust the torque, often with a series of numbers on it. The higher the number, the more torque (rotational force) the drill will apply. If you are having trouble penetrating the material, increase the torque. If you are over-driving screws (burying them too deep), or if drilling too deep could damage something, lower the torque.

4) Avoid overheating the drill bit. If you are drilling through hard materials or drilling at high speeds, the drill bit will encounter an immense amount of friction. This can rapidly overheat the bit, to the point where it becomes red hot or burns the material you're drilling.



Do not force a jammed bit through. If the drill bit gets stuck in the material, don't try to force it out by running the drill. Unplug the drill, separate the bit and the chuck, and remove the bit using manual tools

Do not force a jammed bit through. If the drill bit gets stuck in the material, don't try to force it out by running the drill. Unplug the drill, separate the bit and the chuck, and remove the bit using manual tools

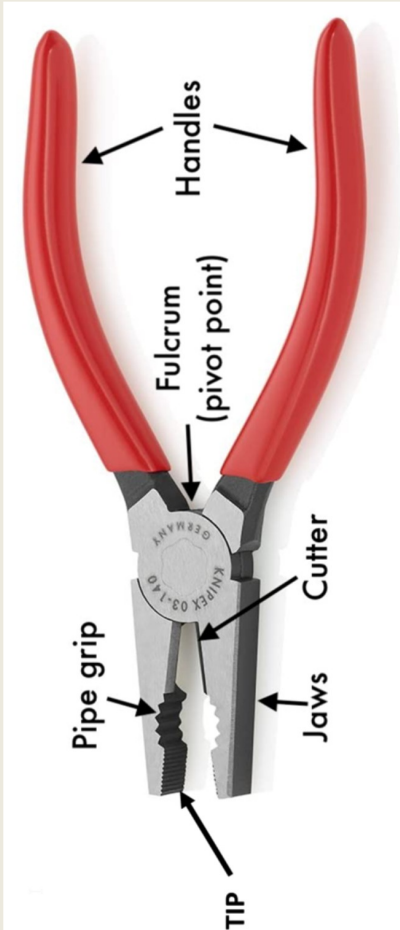


5) Chamfer the holes both sides to remove all sharp edges using a bigger size drill bit or countersink tool



countersink tool

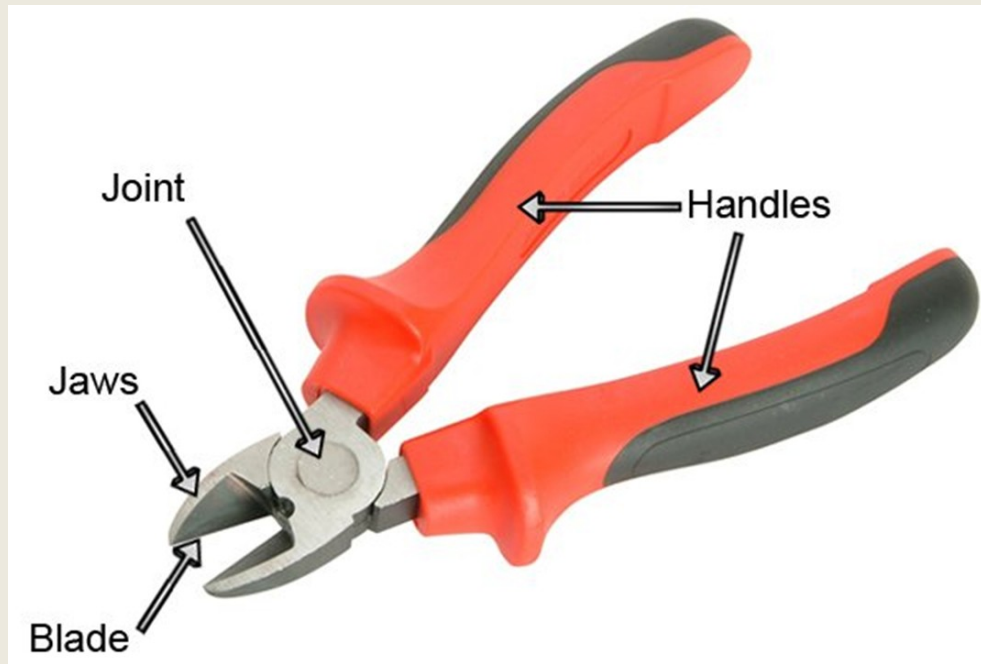
General Pliers



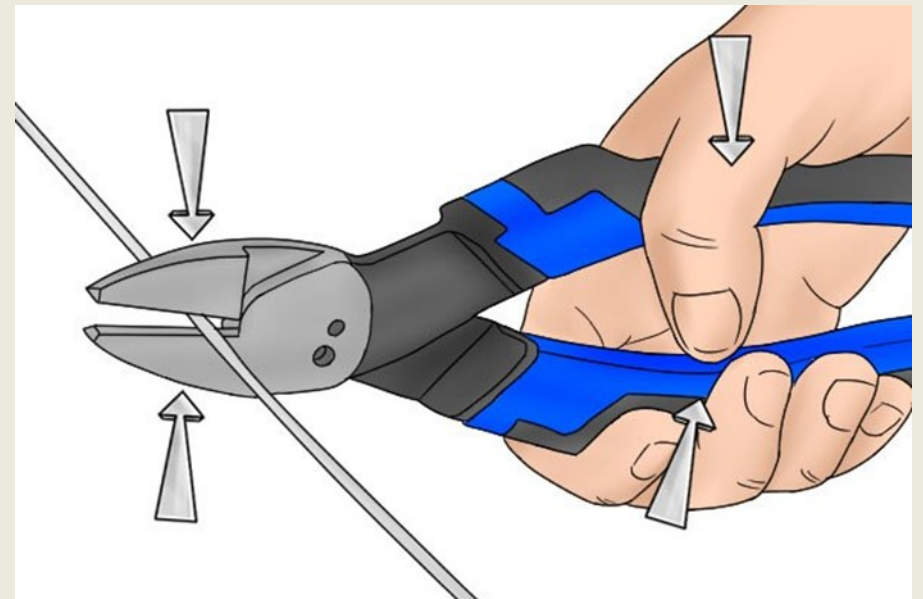
- Tip: The tip is used to pinch or hold an object with precision.
- Pipe Grip: The rounded shape of the pipe grip allows you to hold on tight to rounded objects, such as pipes.
- Jaws: The jaws has a specific size and determines the size of the pliers.
- Fulcrum: This is the pivot point around which the parts of the pliers rotate.

Some pliers also have a grab or pinching point just behind the fulcrum.

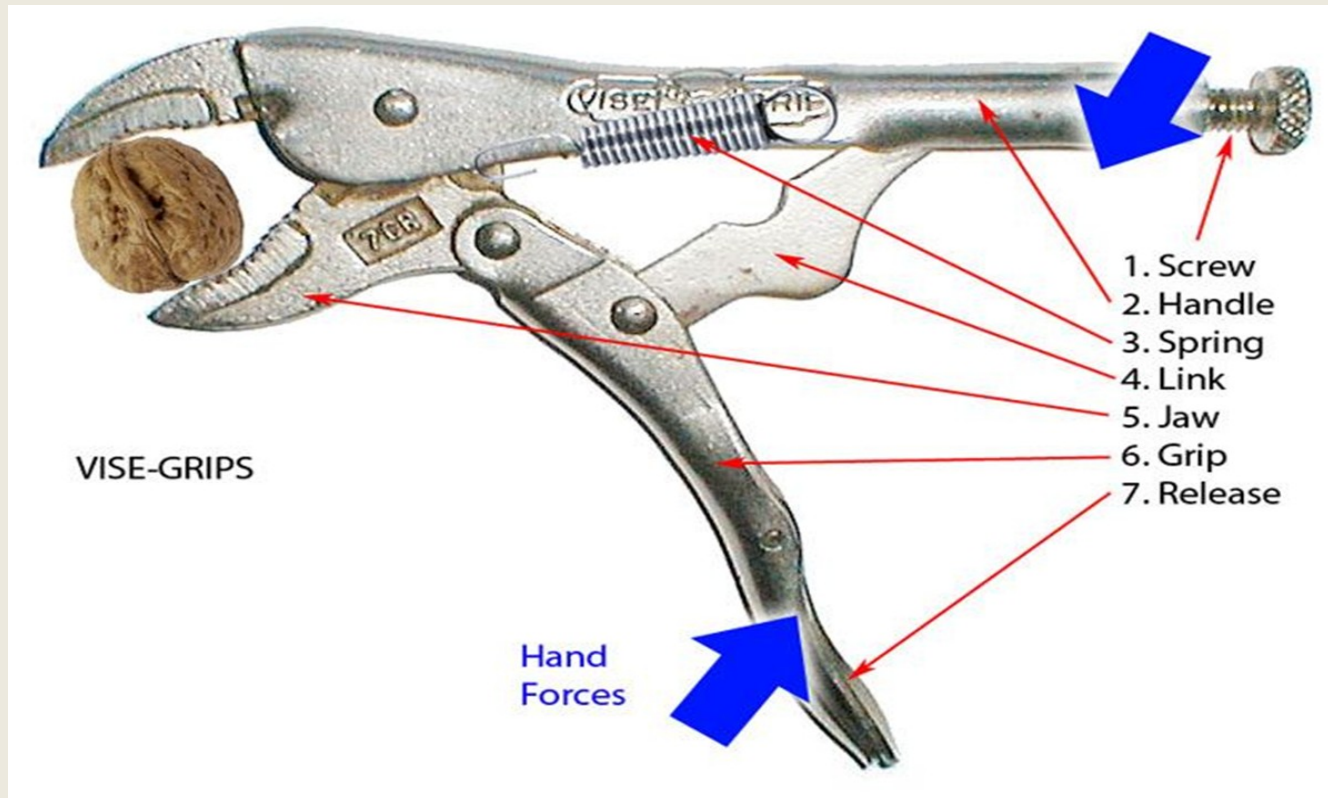
- Handles: These are used to hold onto the pliers. Some pliers have insulated handles so the pliers can be used on electrical wiring.



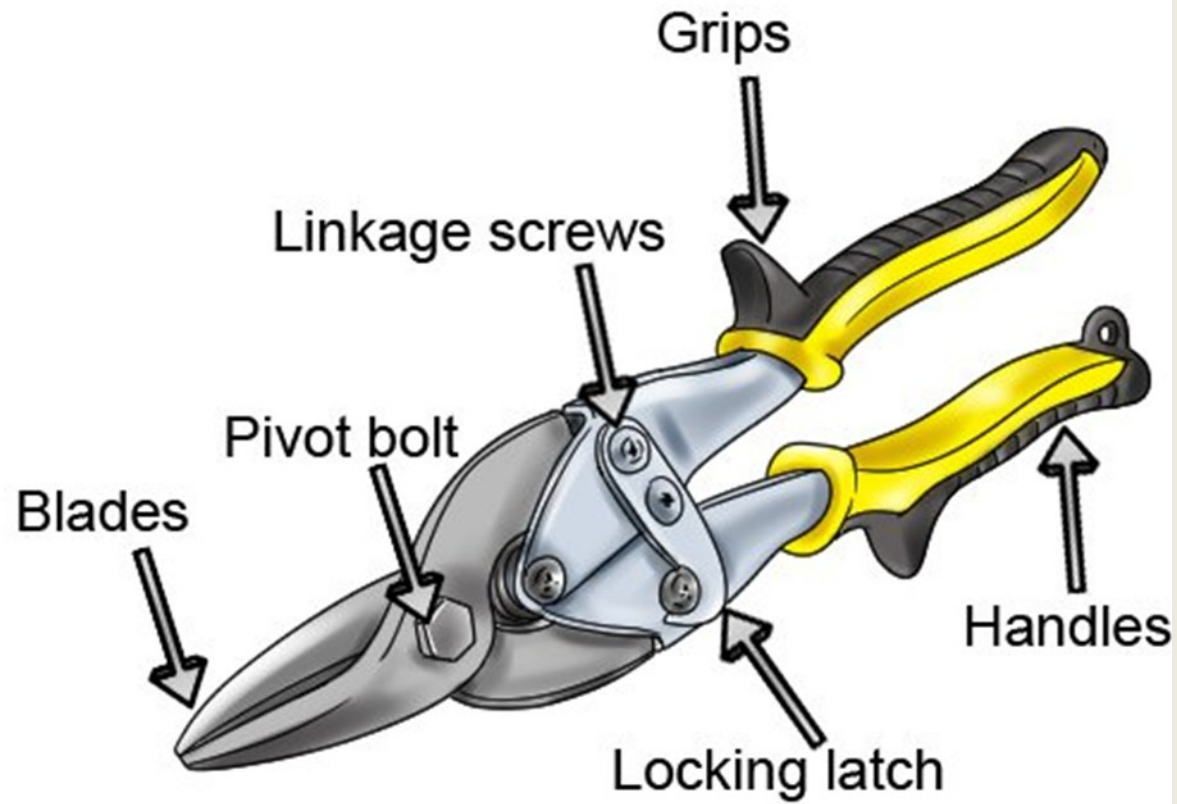
Side Cutter



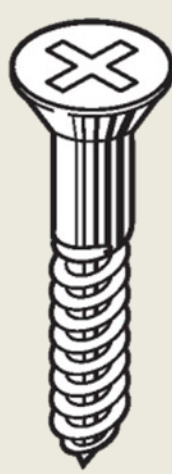
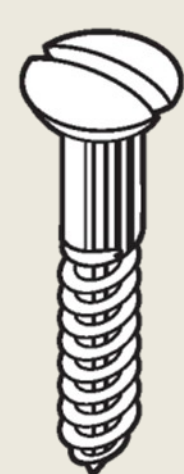
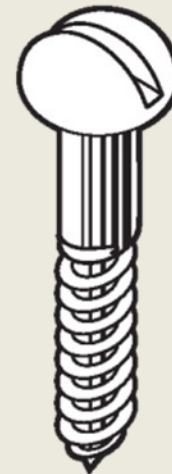
VICE GRIP



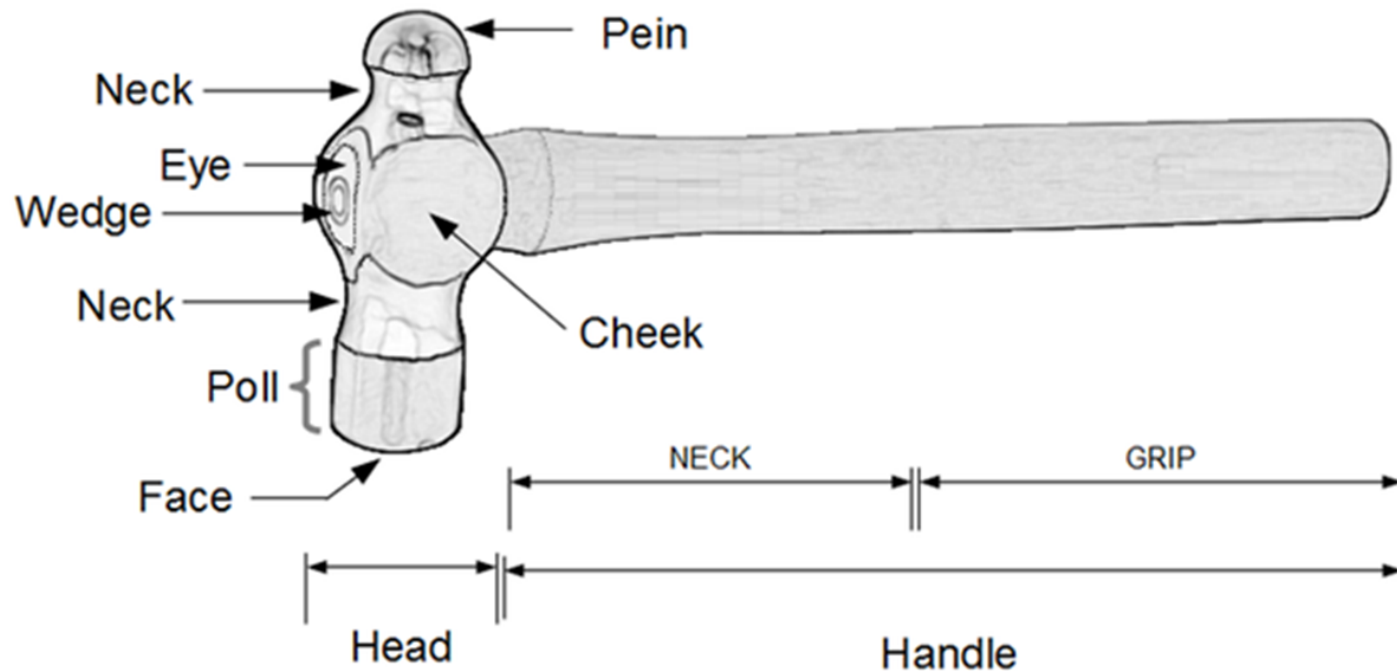
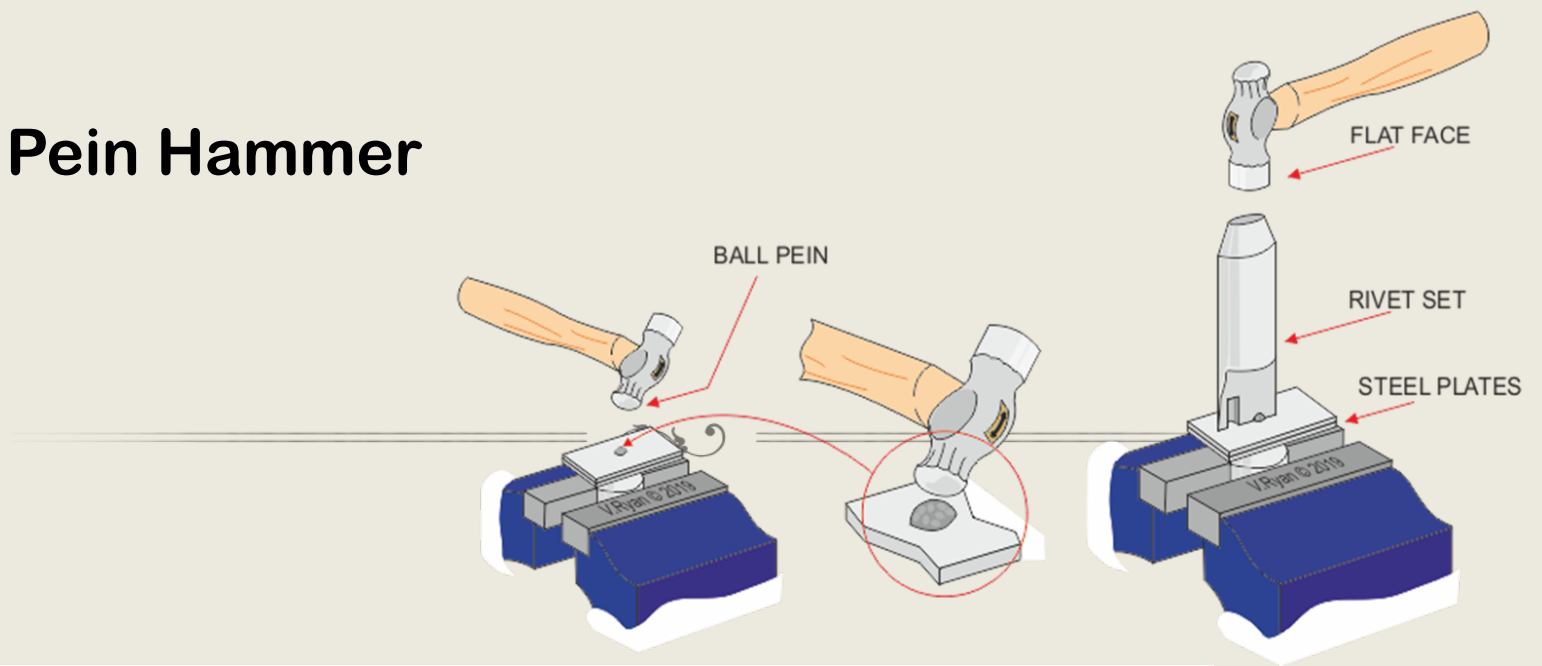
Tin snips



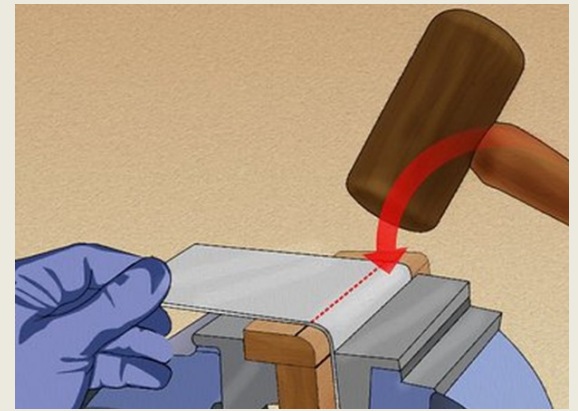
SCREW DRIVERS



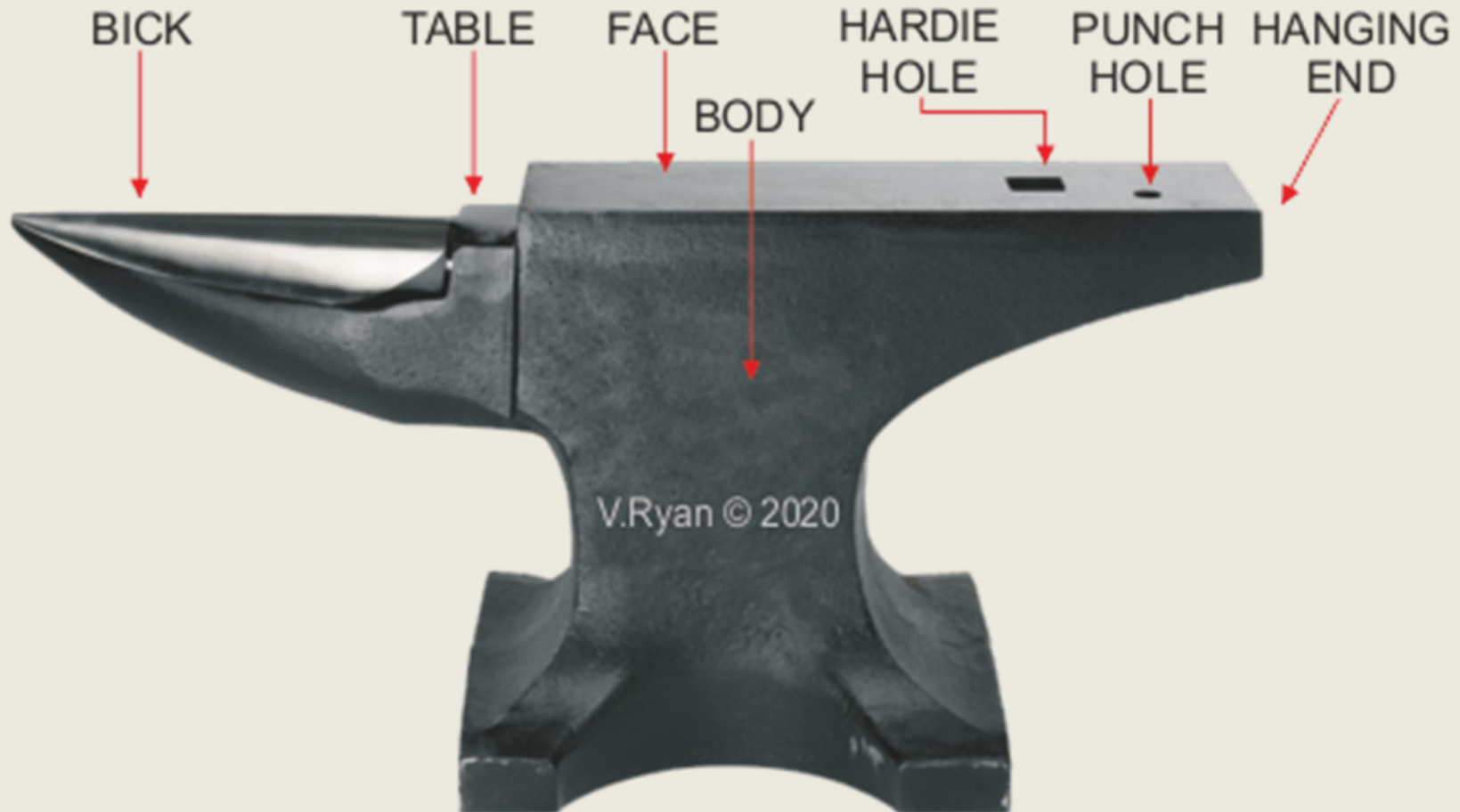
Ball Pein Hammer



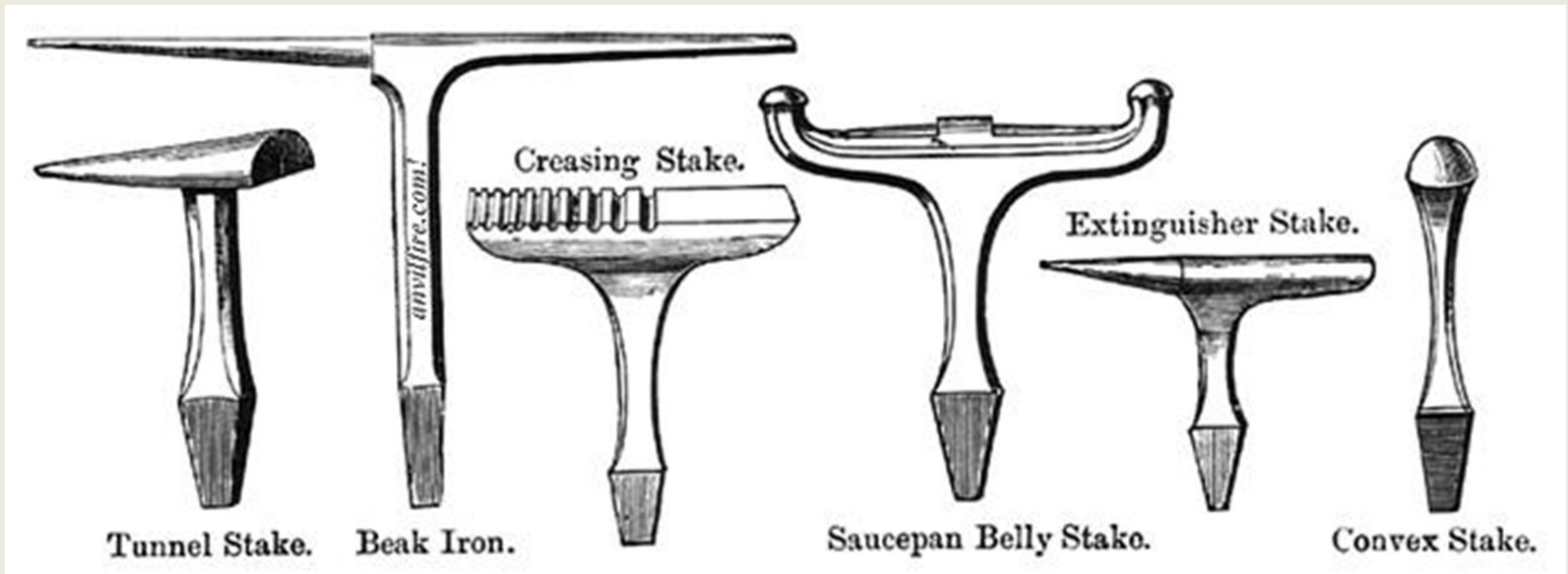
MALLET



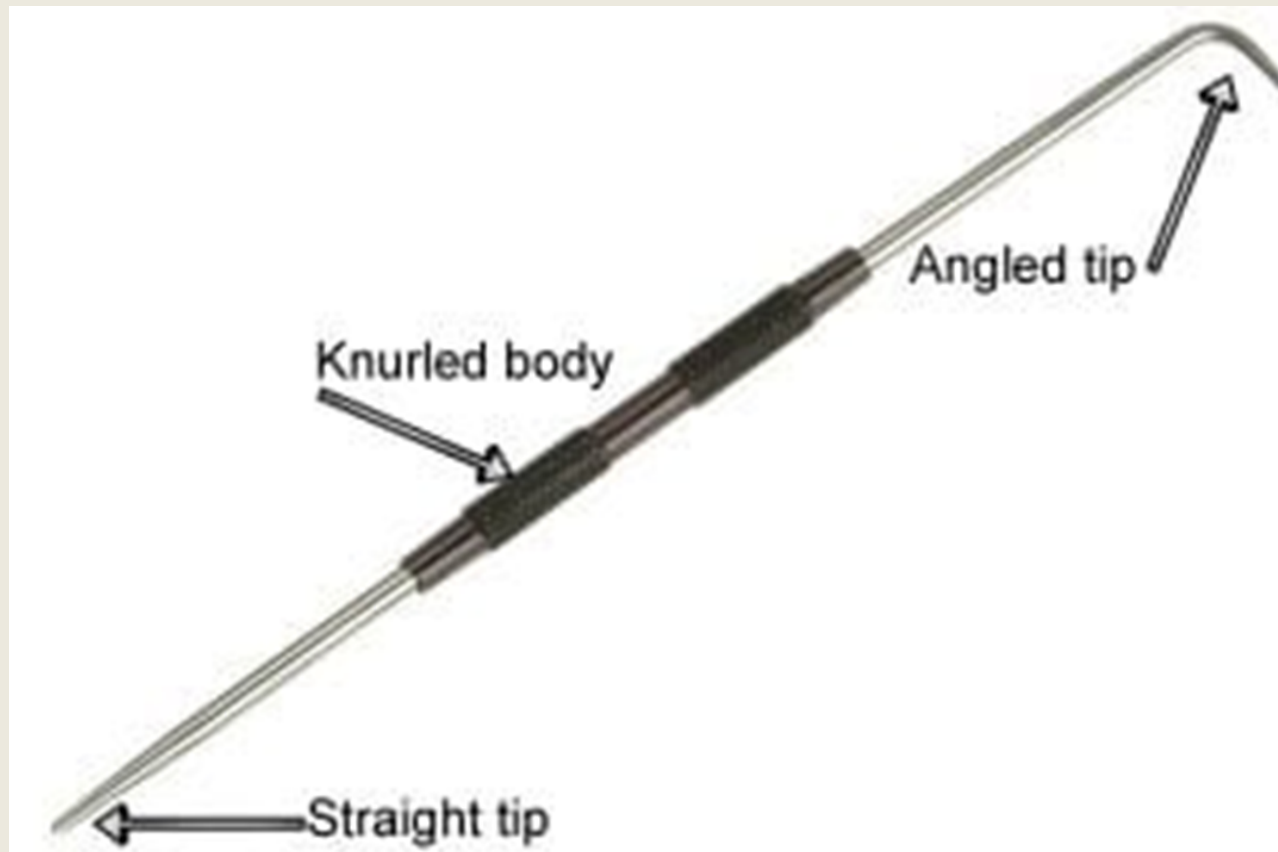
THE ANVIL



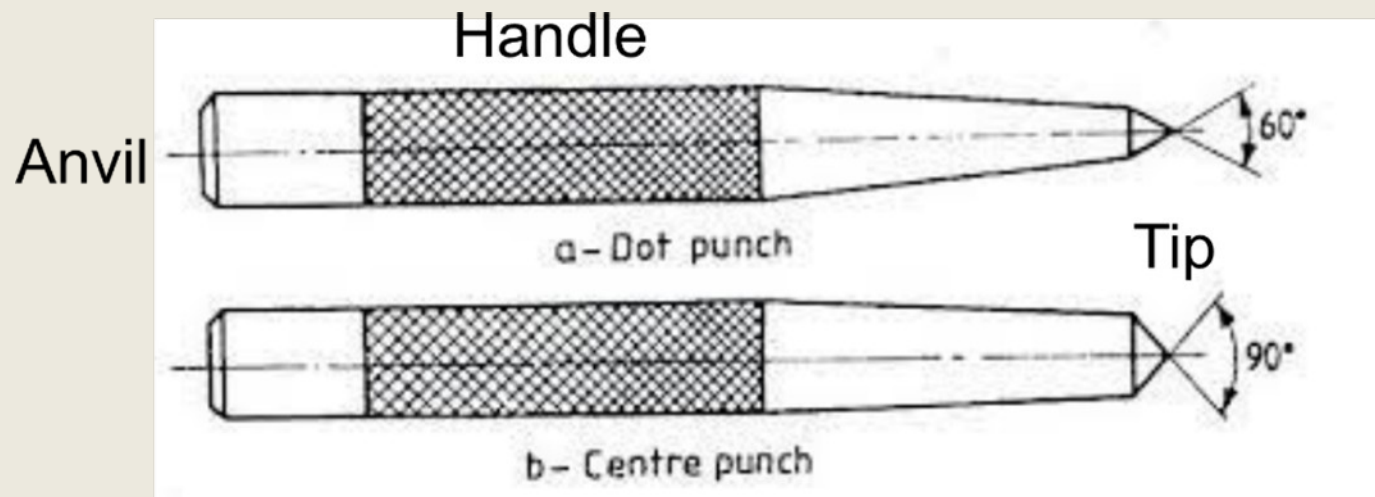
STAKES

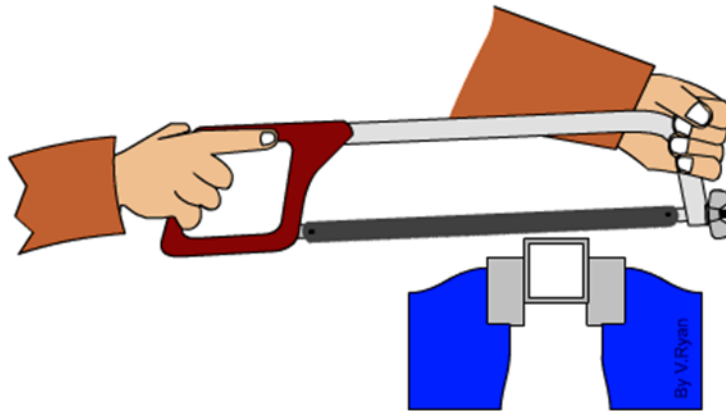
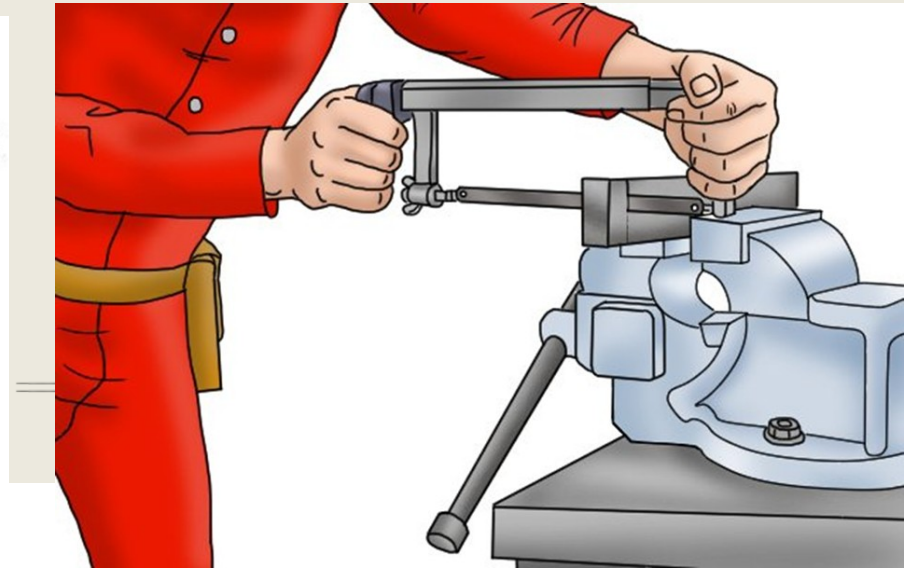
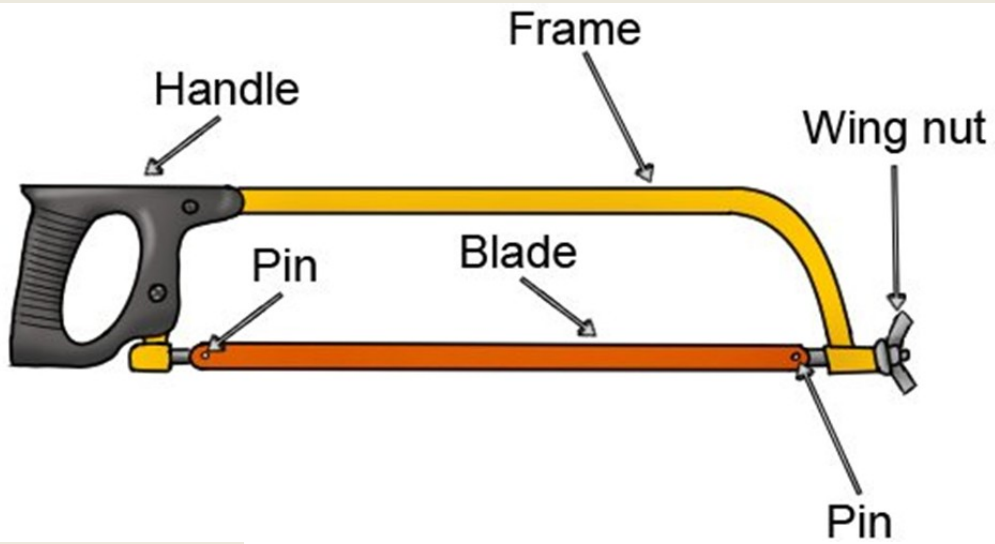


THE SCRIBE

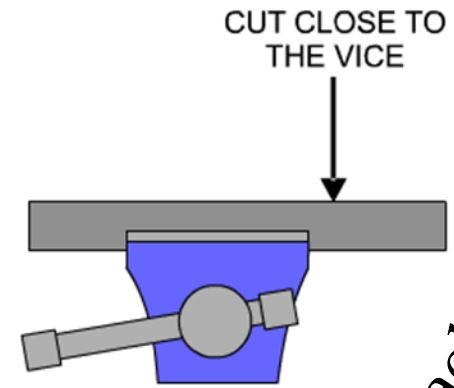


PUNCH





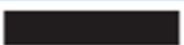
SIDE VIEW

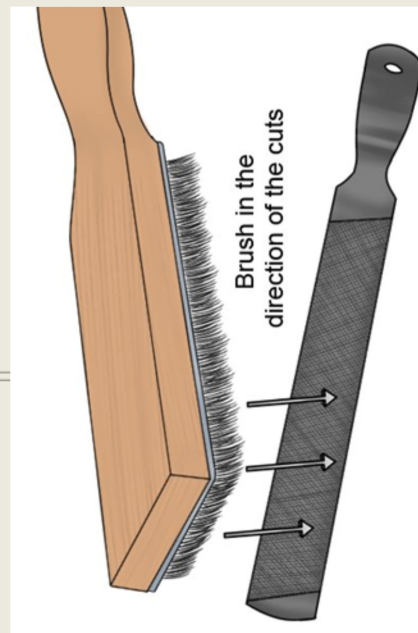


FRONT VIEW

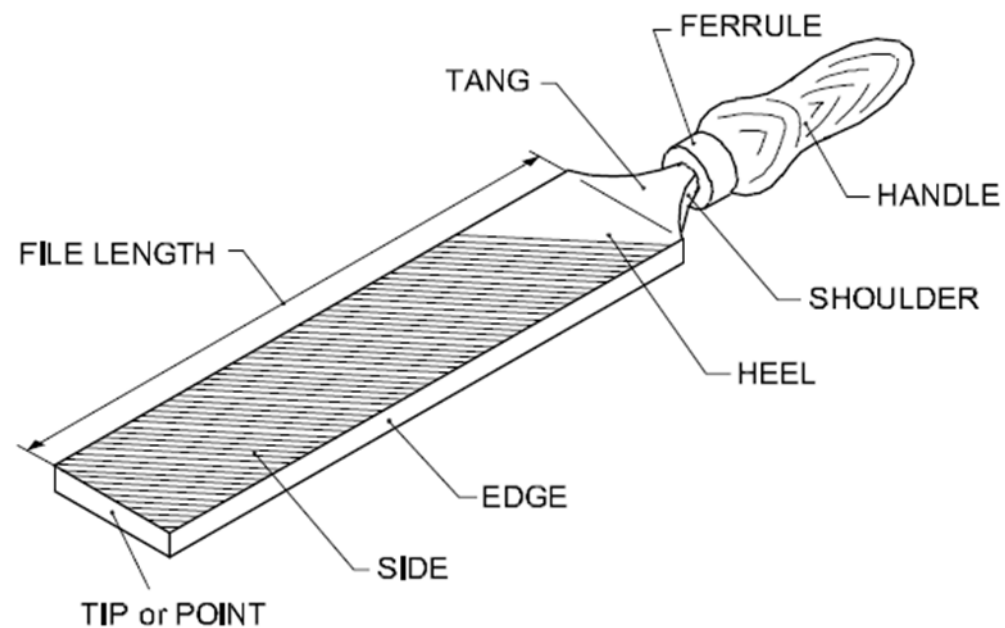
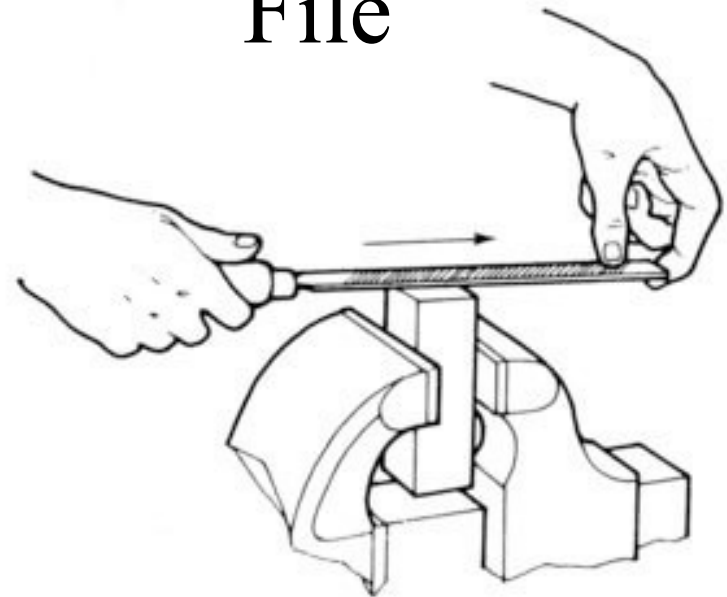
Hacksaw

FILE PROFILES

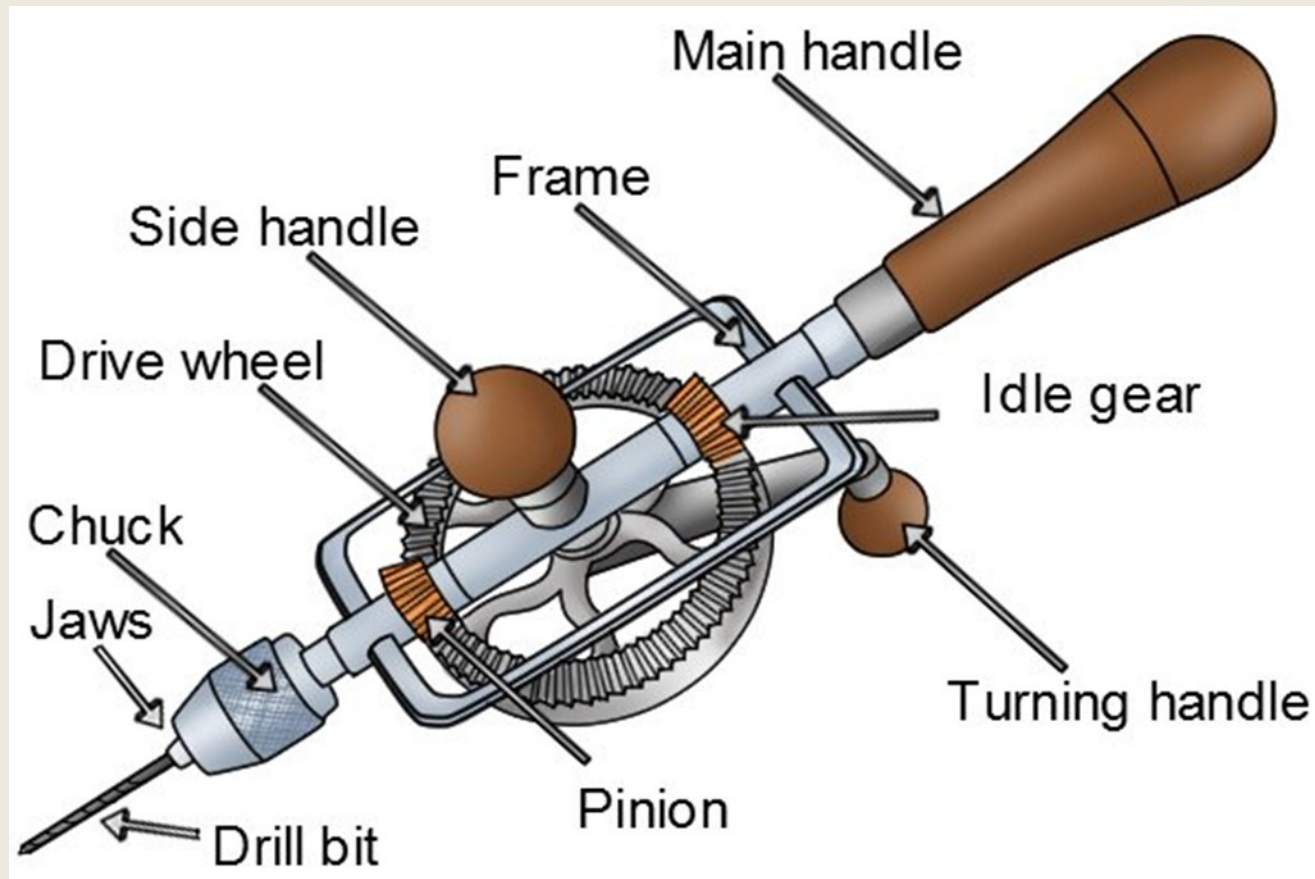
Profile	Use
	Smoothing large curves or flat areas
	Roughing out or shaping large inside curves
	Tight areas or hard to reach spots
	Shaping or enlarging holes, or shaping tight inside curves
	Shaping inside corners
	Smoothing inside curves
	Tight areas where clearance is a problem



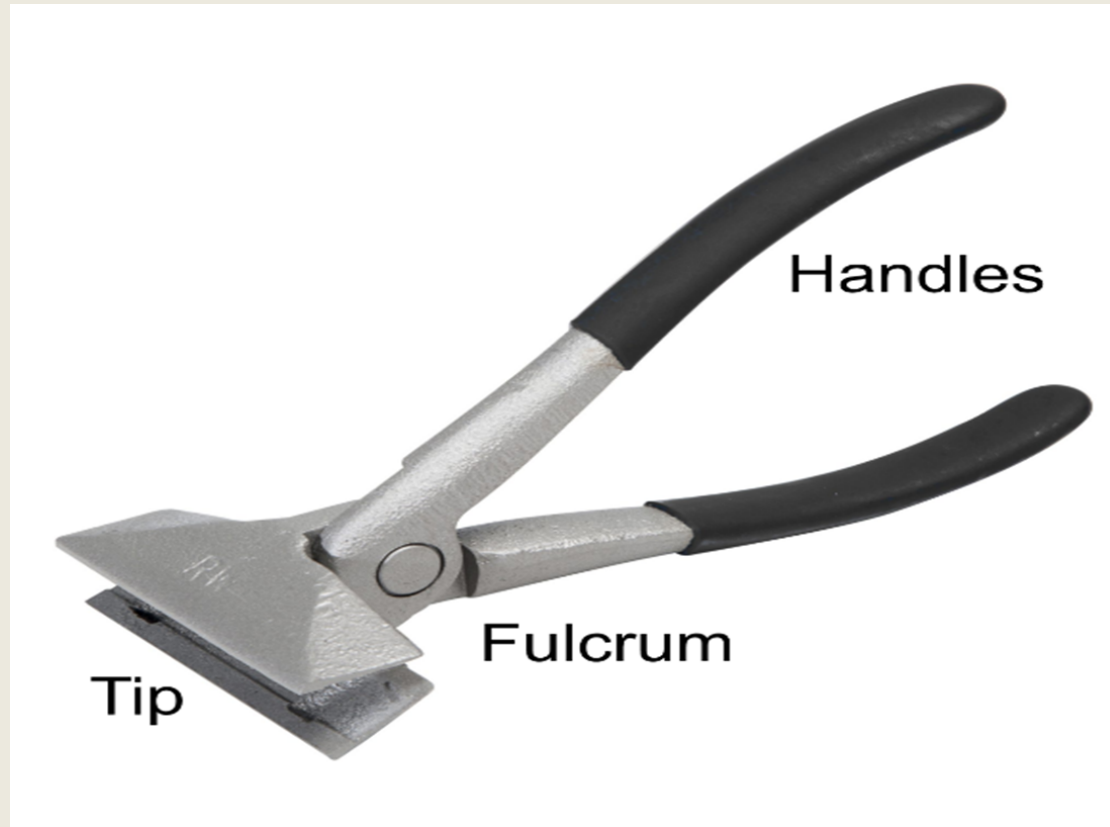
File



HAND DRILL



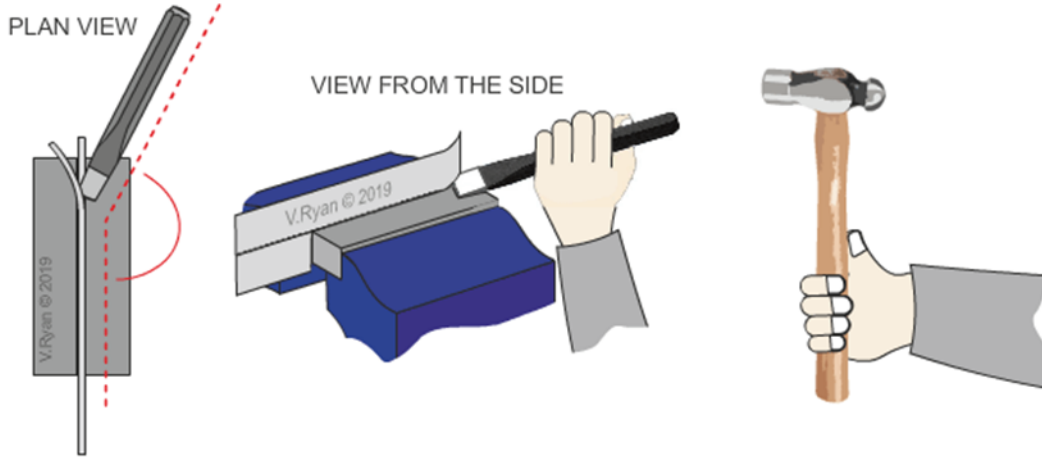
HAND SEAMER



Cold Chisel

PLAN VIEW

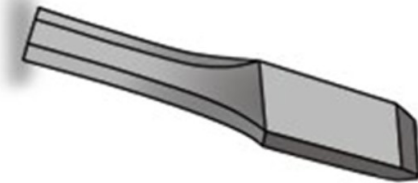
VIEW FROM THE SIDE



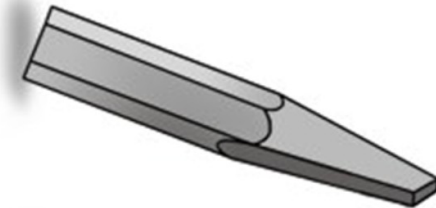
Diamond
Point
chisel



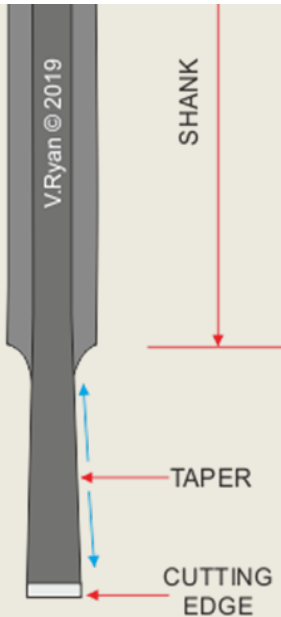
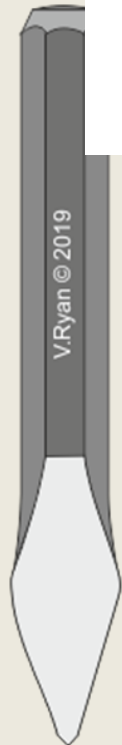
Flat
chisel



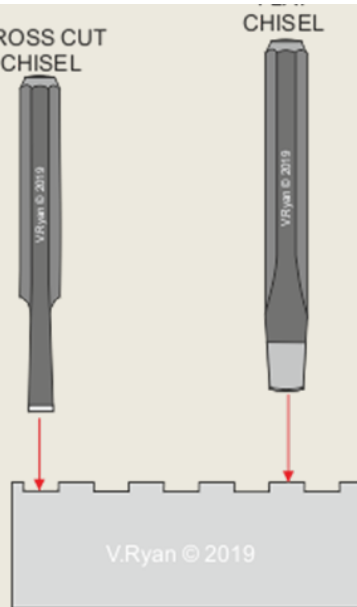
Half
round
chisel



Cross cut/
cape
chisel



CROSS CUT
CHISEL

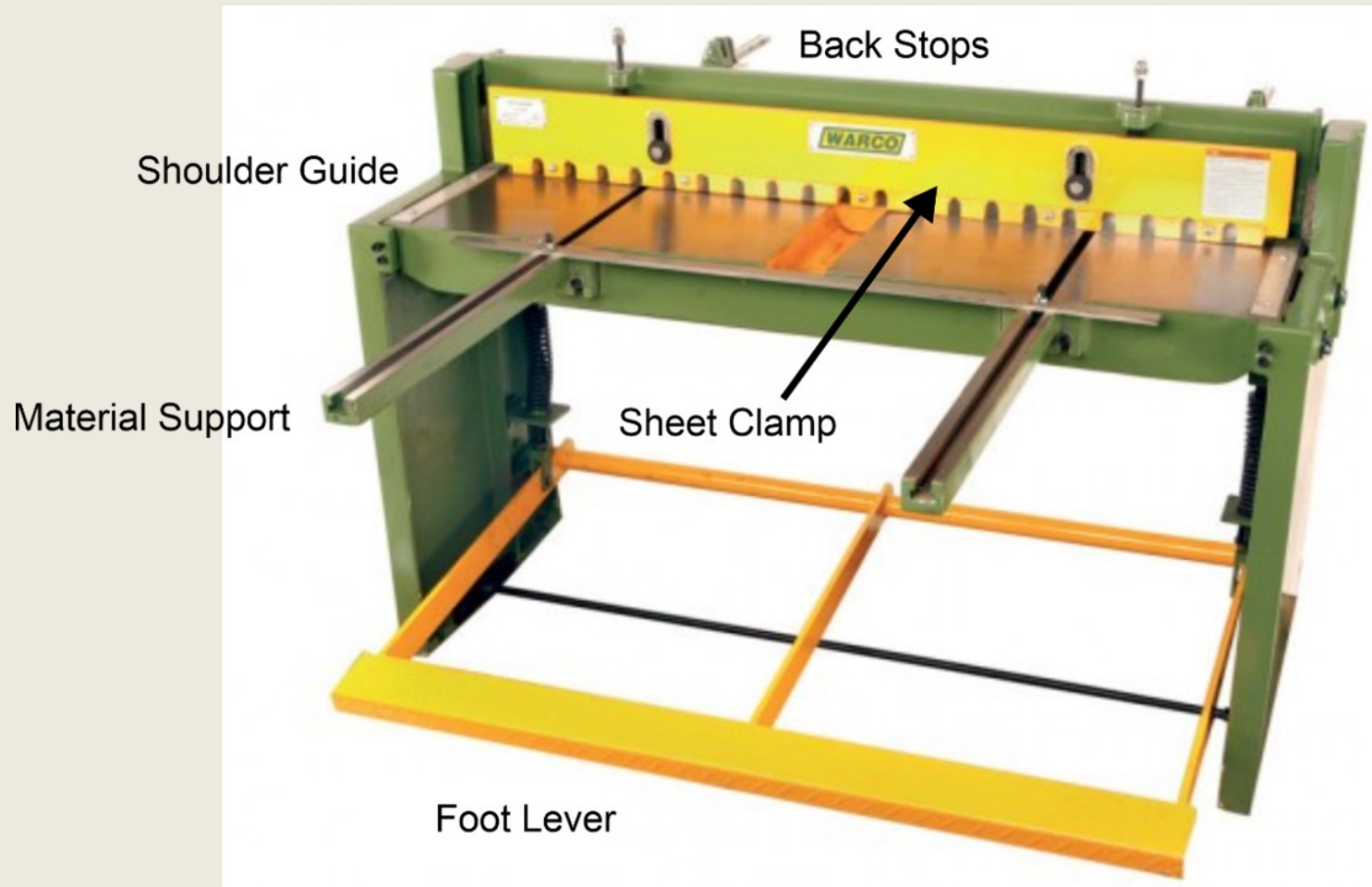


CHISEL

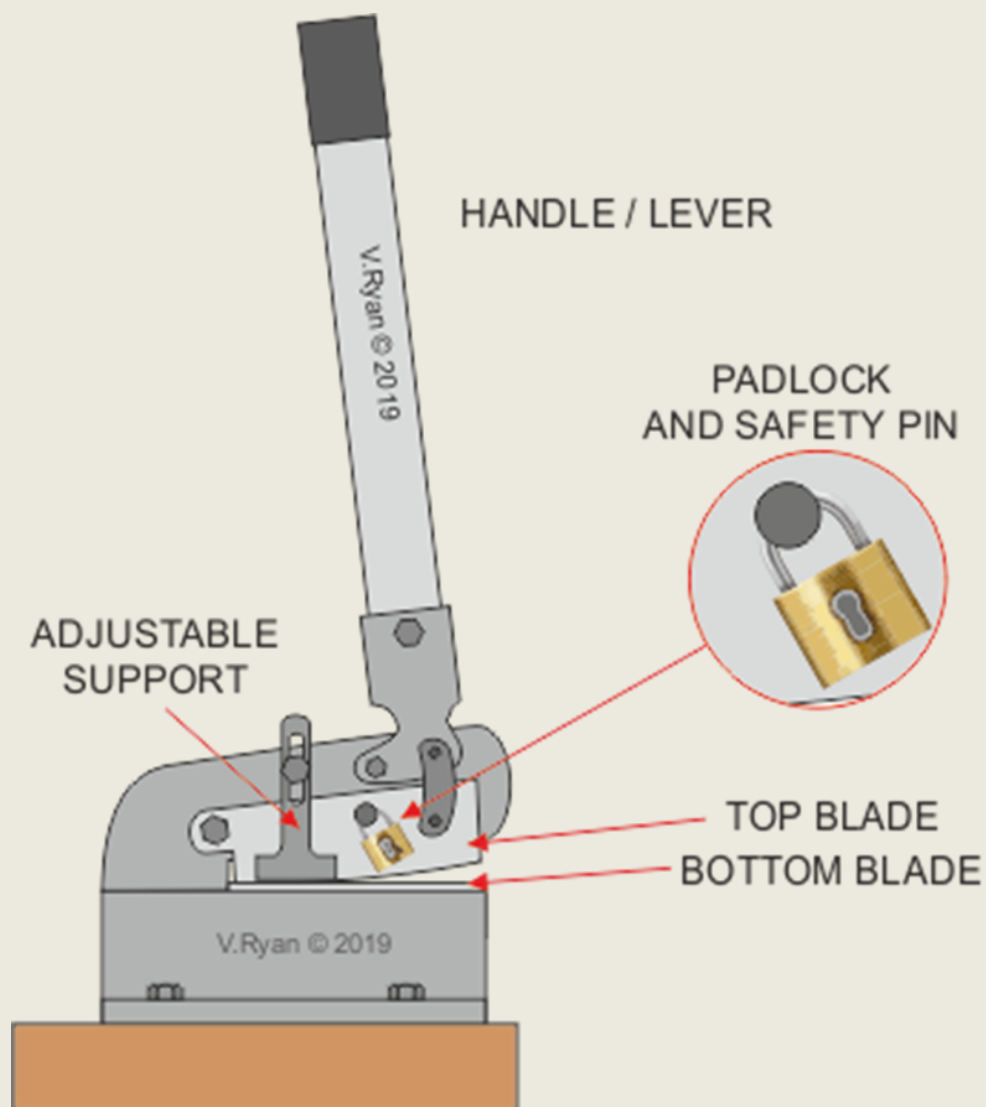


V.Ryan © 2019

Guillotine: Foot Operated



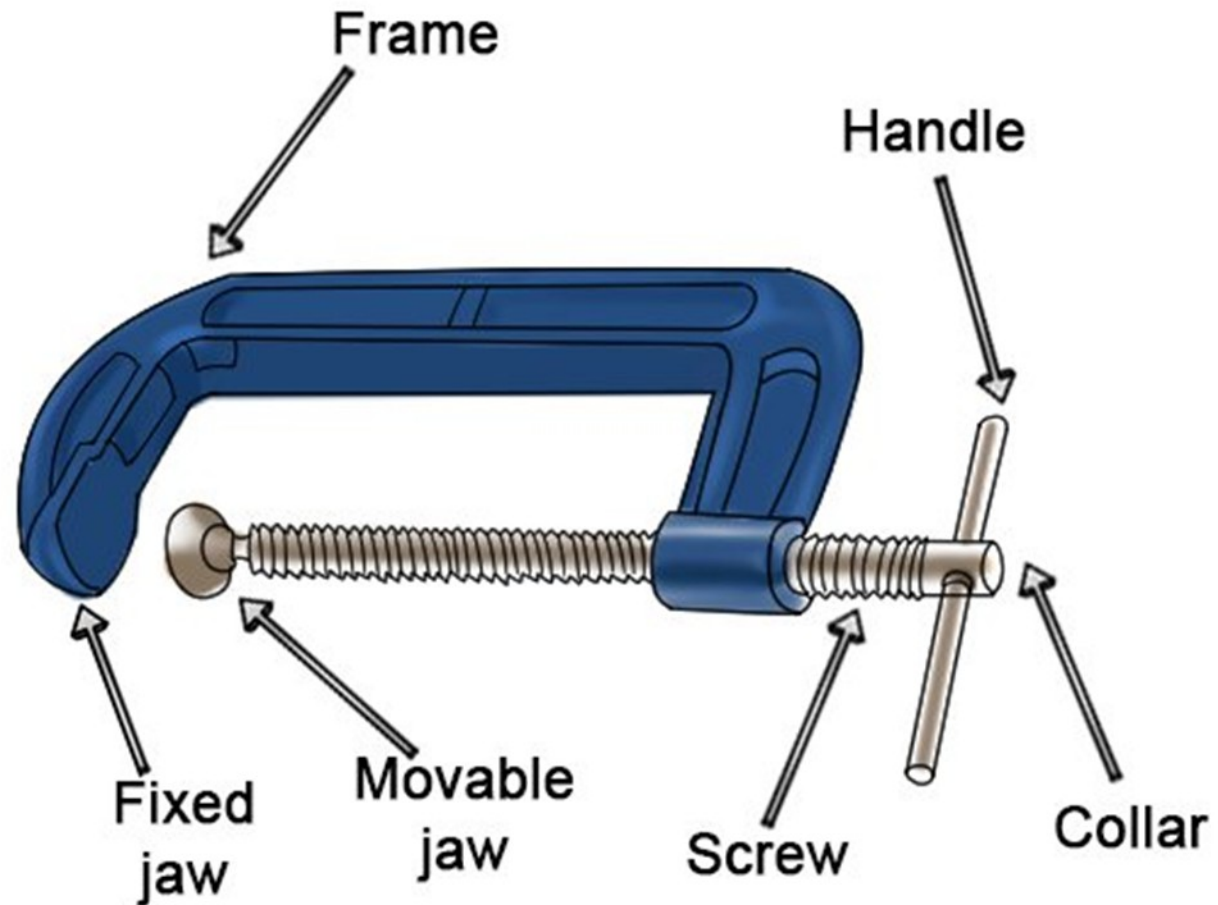
Hand Guillotine / Shear



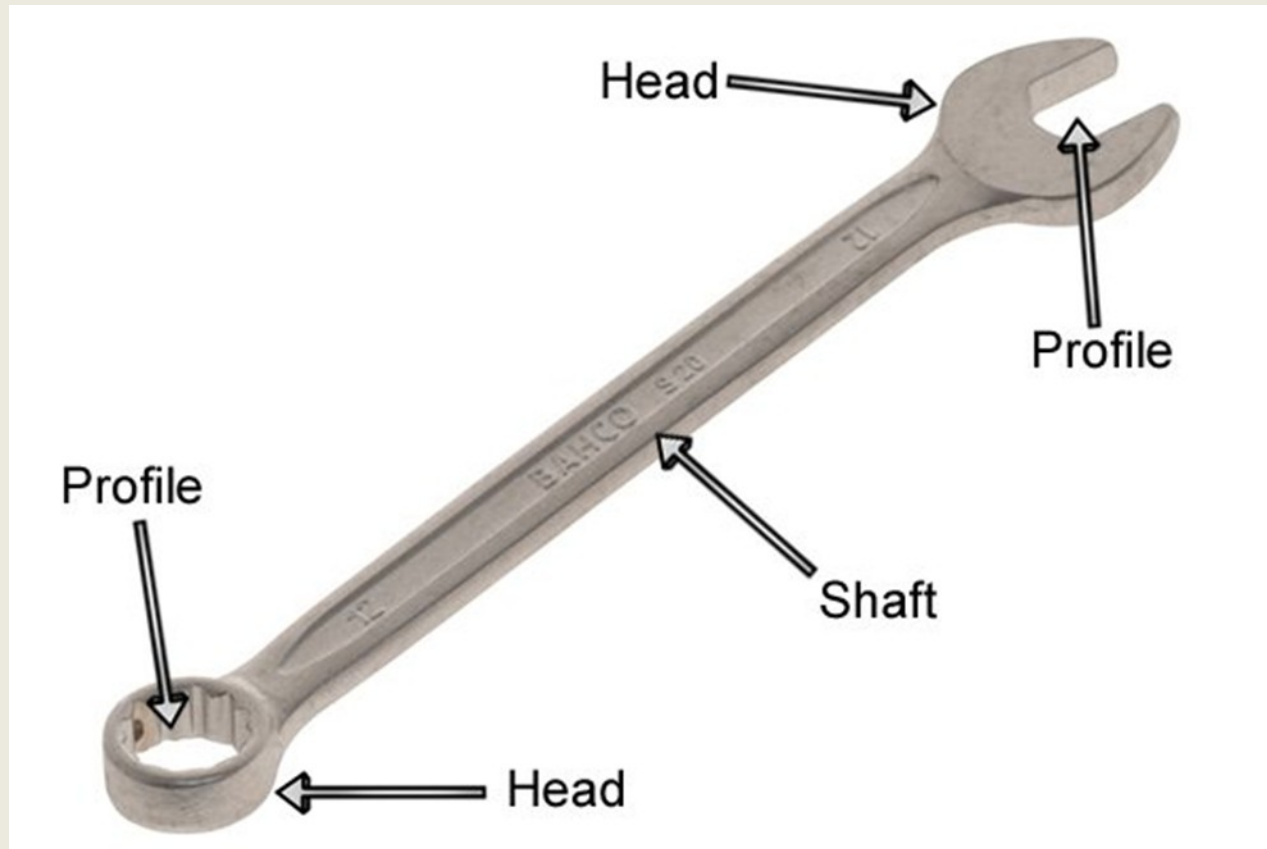
Nibbler



G Clamps



JOINING TOOLS: SPANNERS



JIGS

Jigs are used when a worker needs to manufacture many parts that are the same.

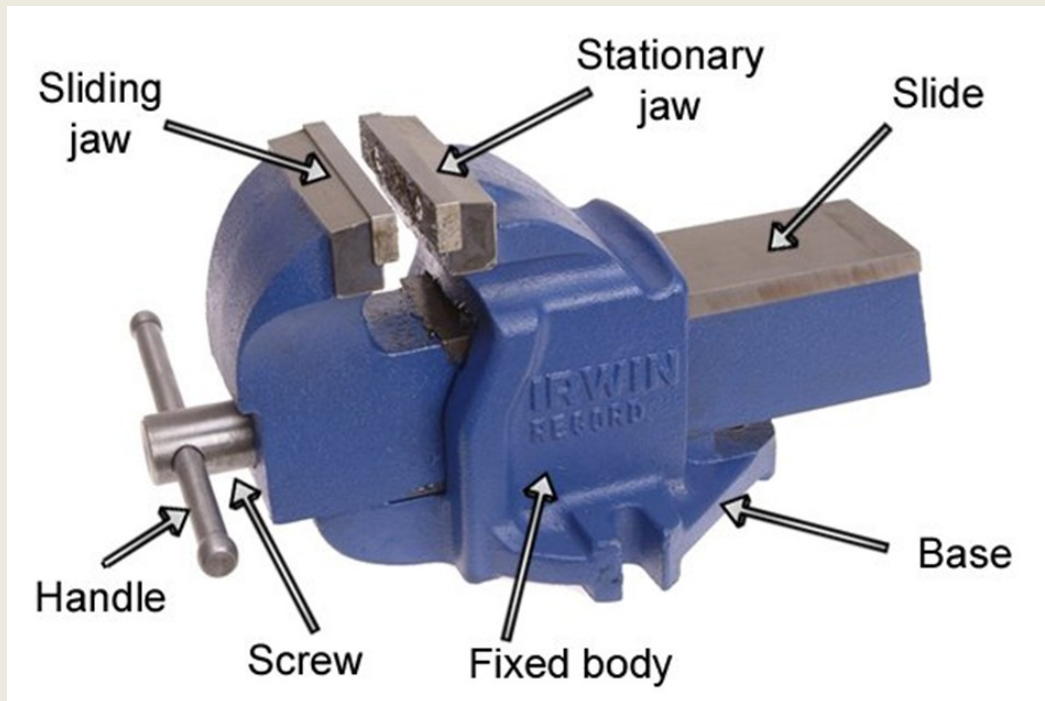
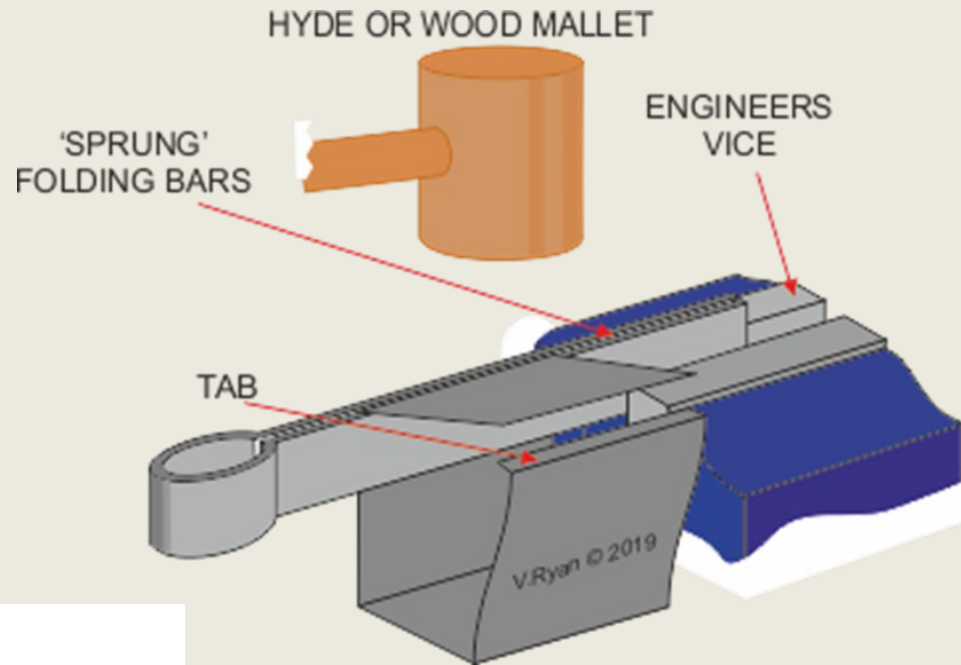
When making similar parts, you will do the same measurements and materials handling time after time.

One way to cut down on manufacturing time and ensuring uniformity in manufacturing is utilising a jig.

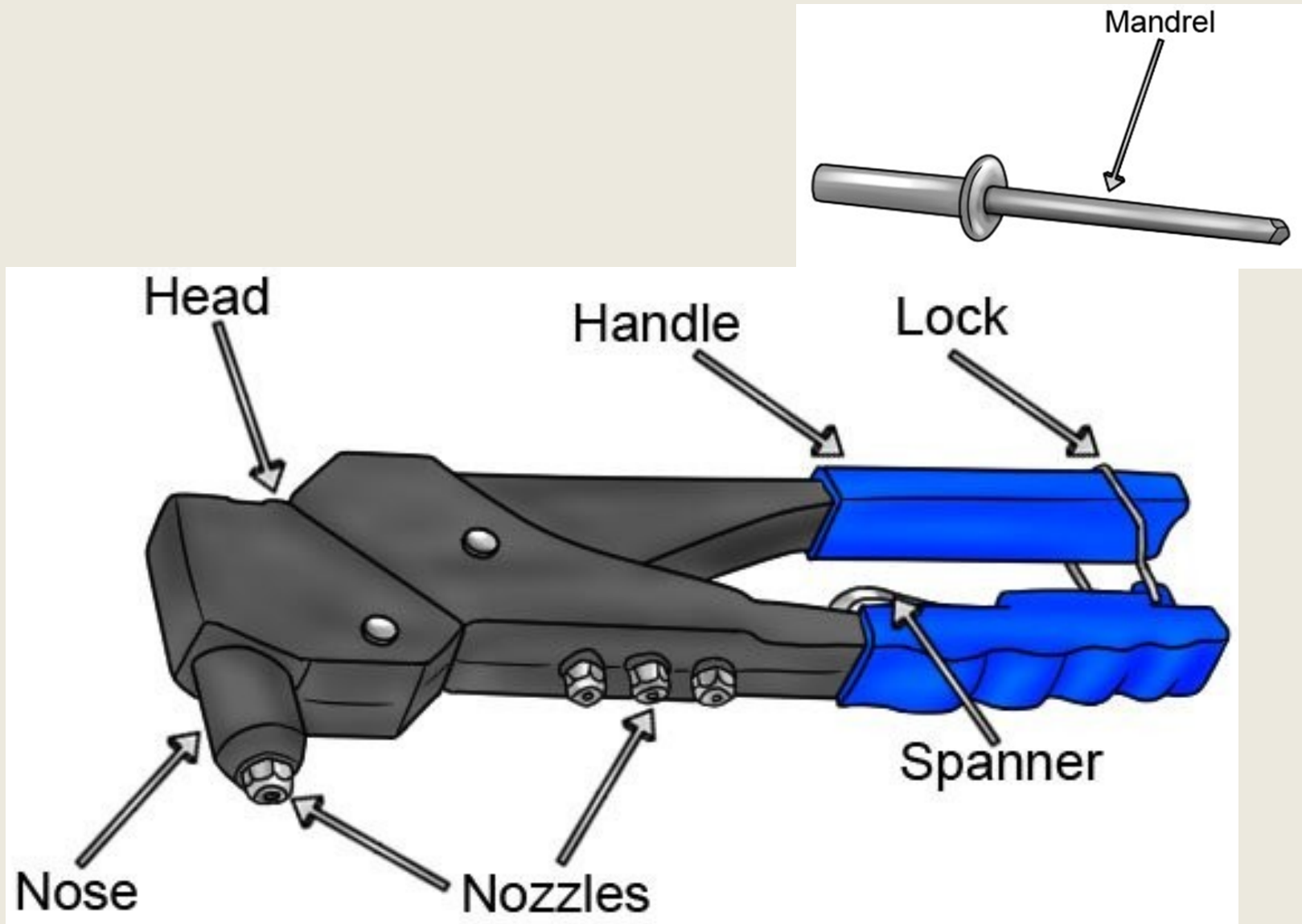
A jig is a pre-formed clamp or positioning points that ensure that the workpiece can be place in the same position time after time, making the same cut or bend time after time



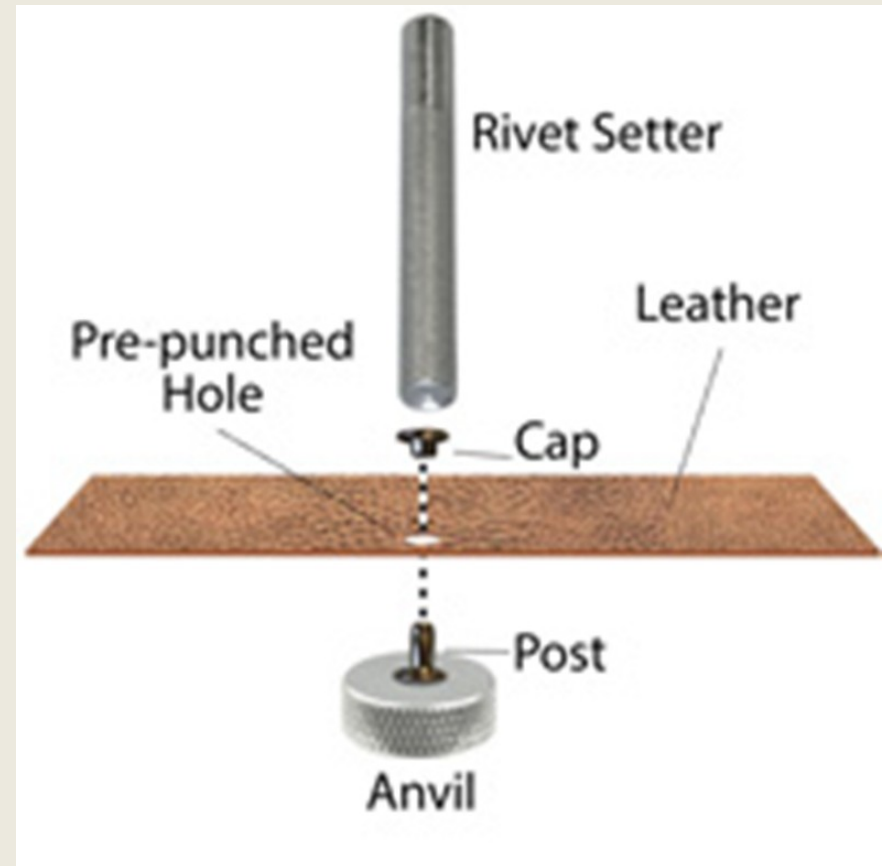
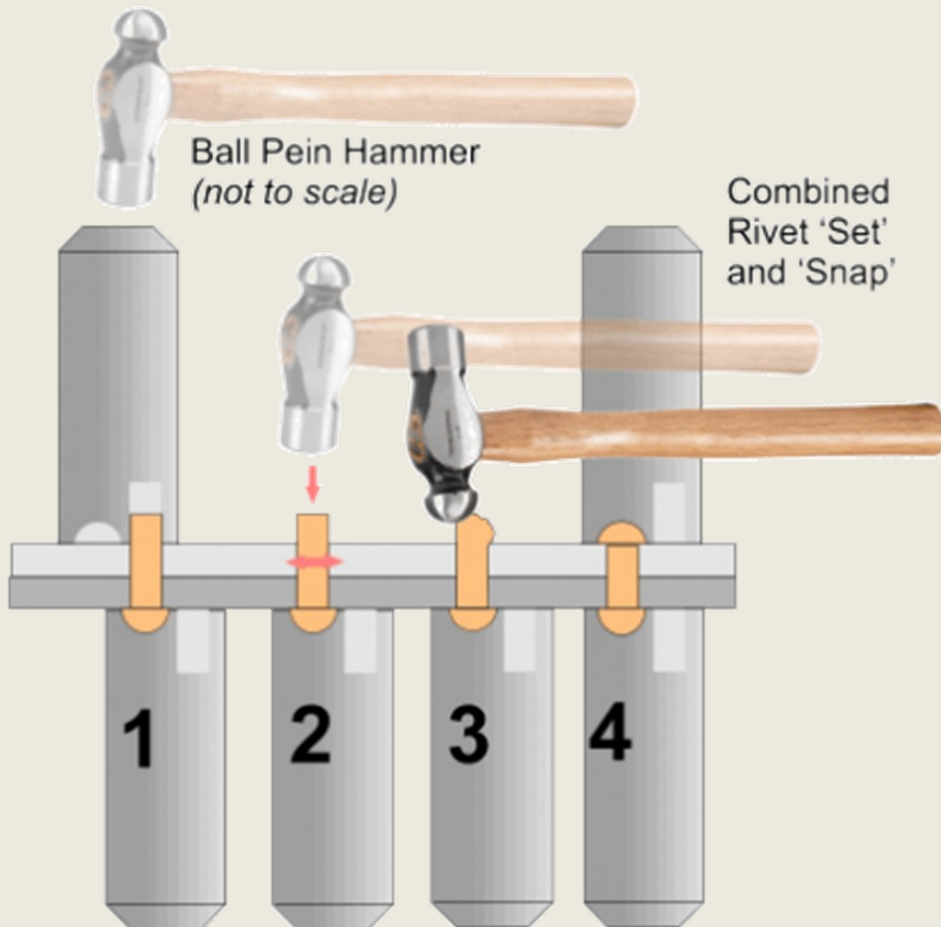
Engineers Vice & Folding Bars



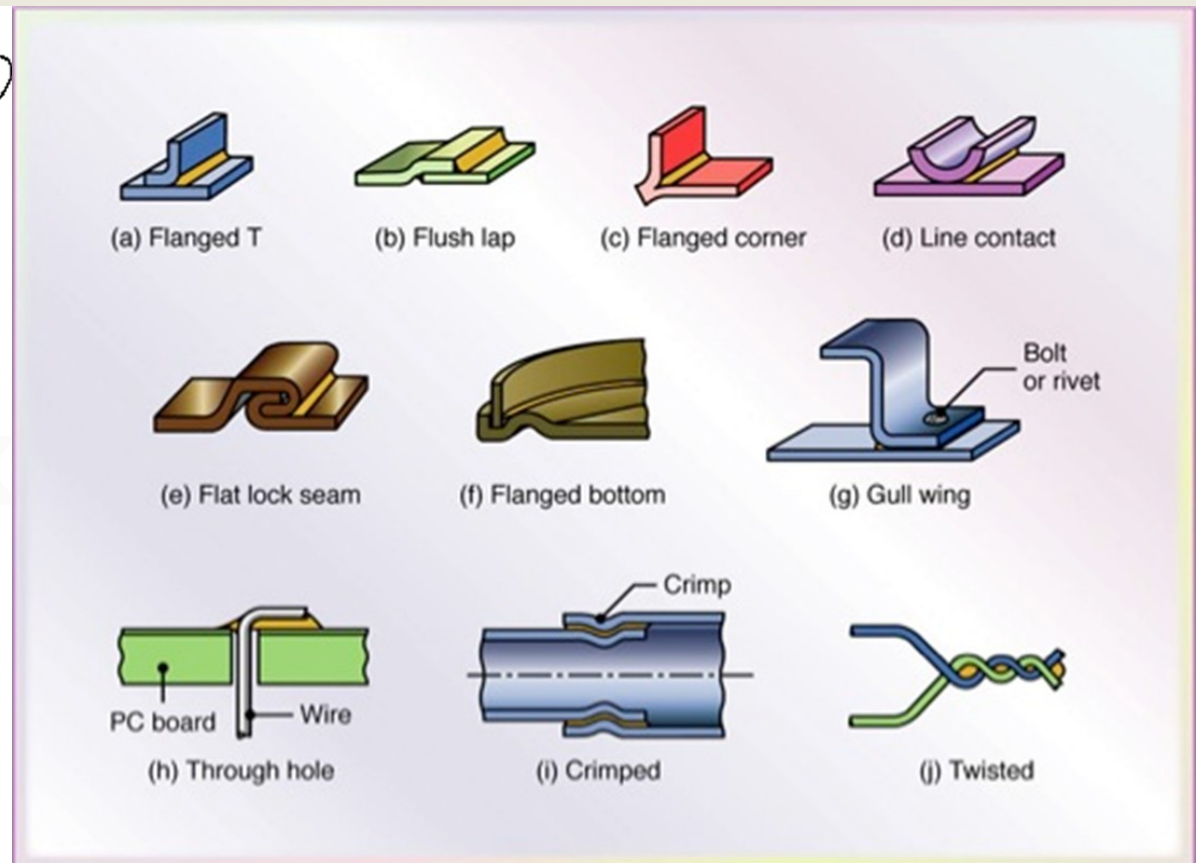
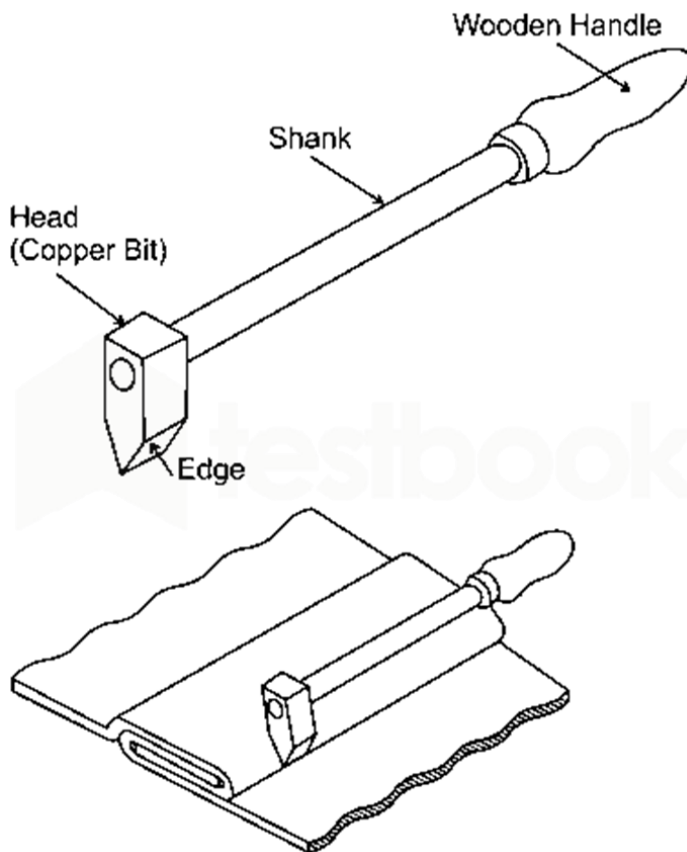
Joining Tools: Rivet gun



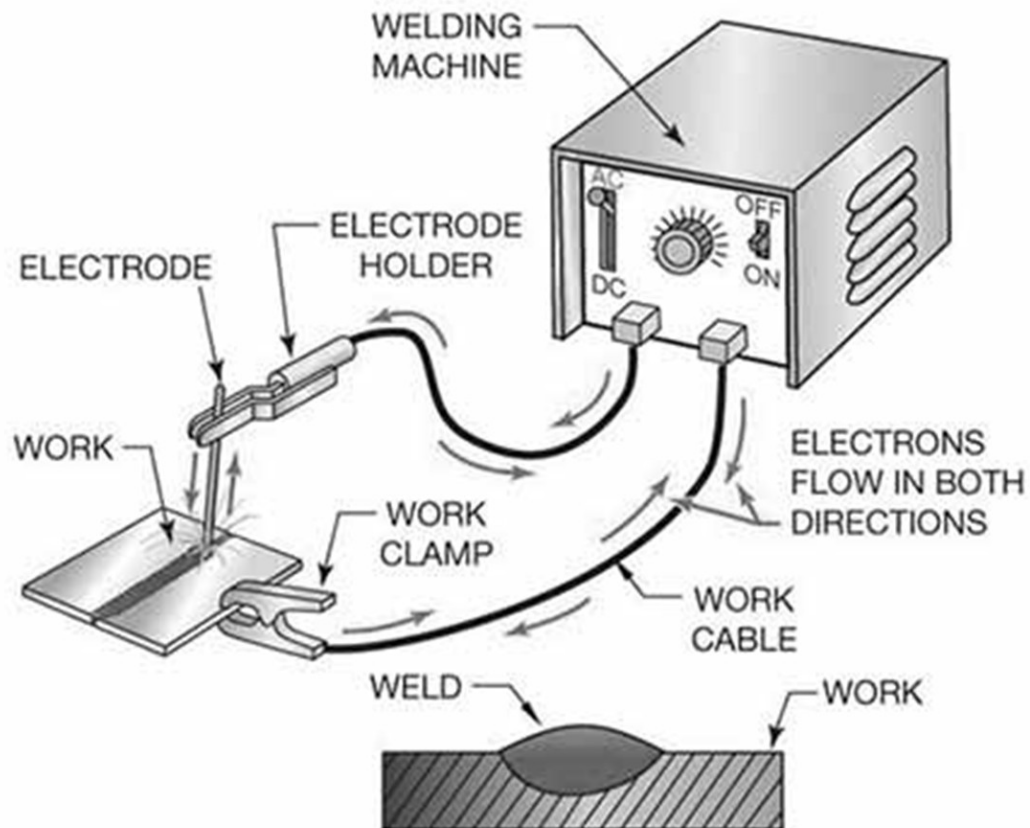
Joining Tools: Rivet Snap Set



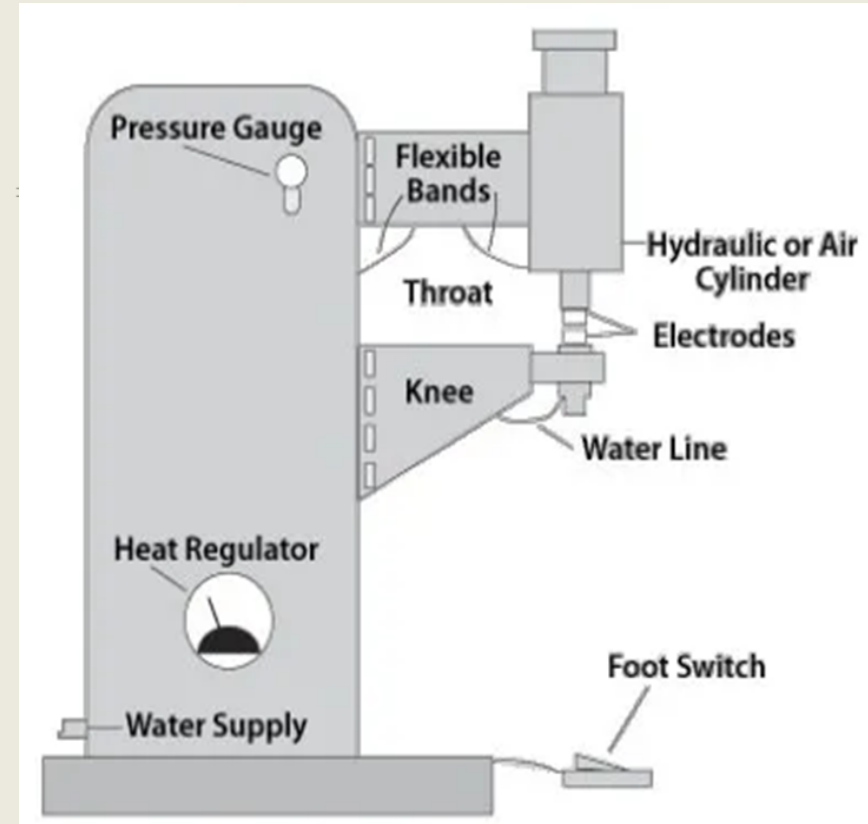
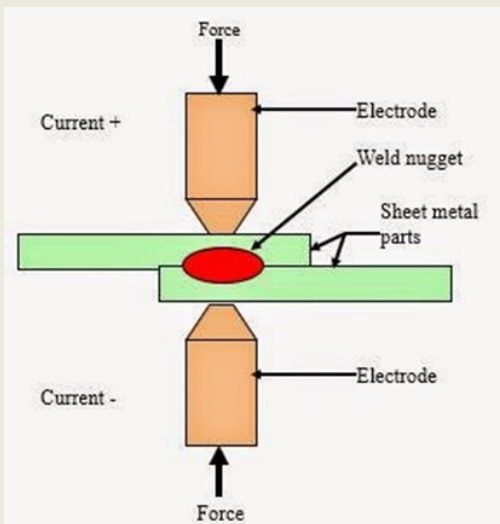
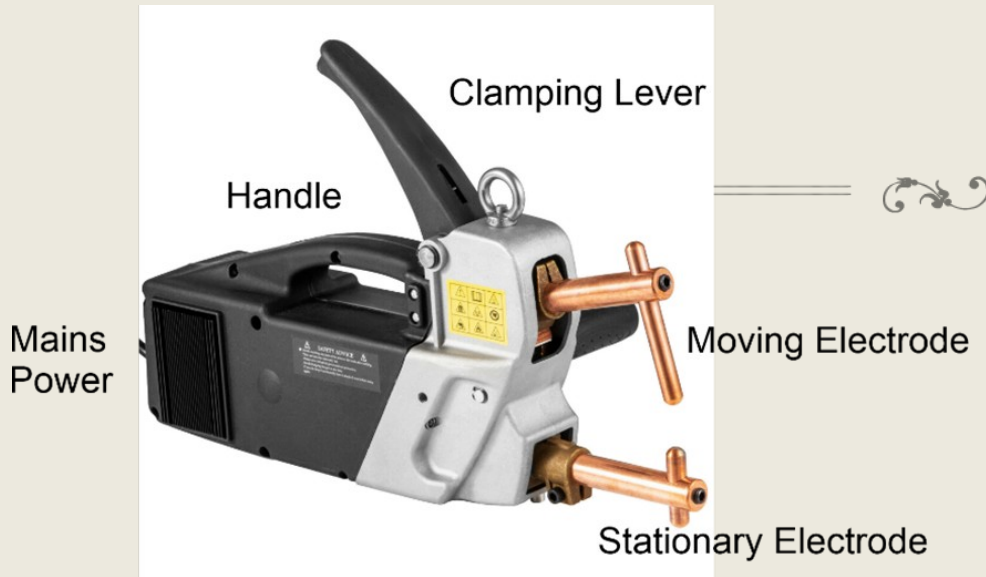
Joining Tools: Solder Iron



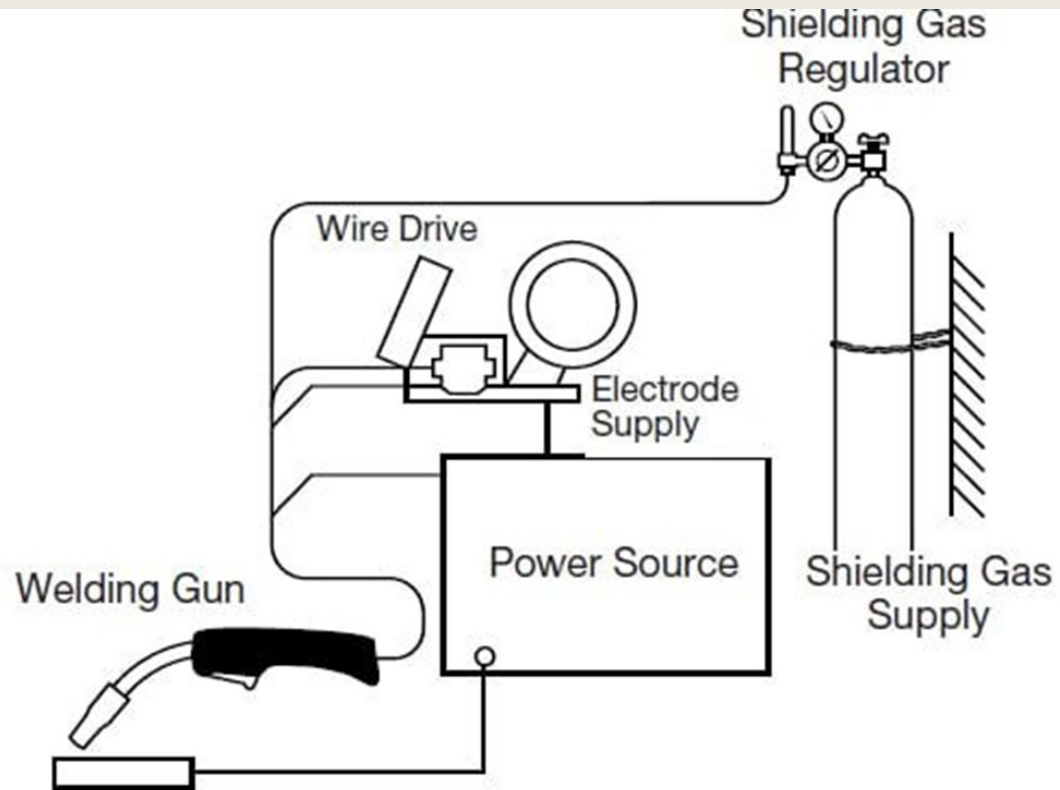
Stick Welder



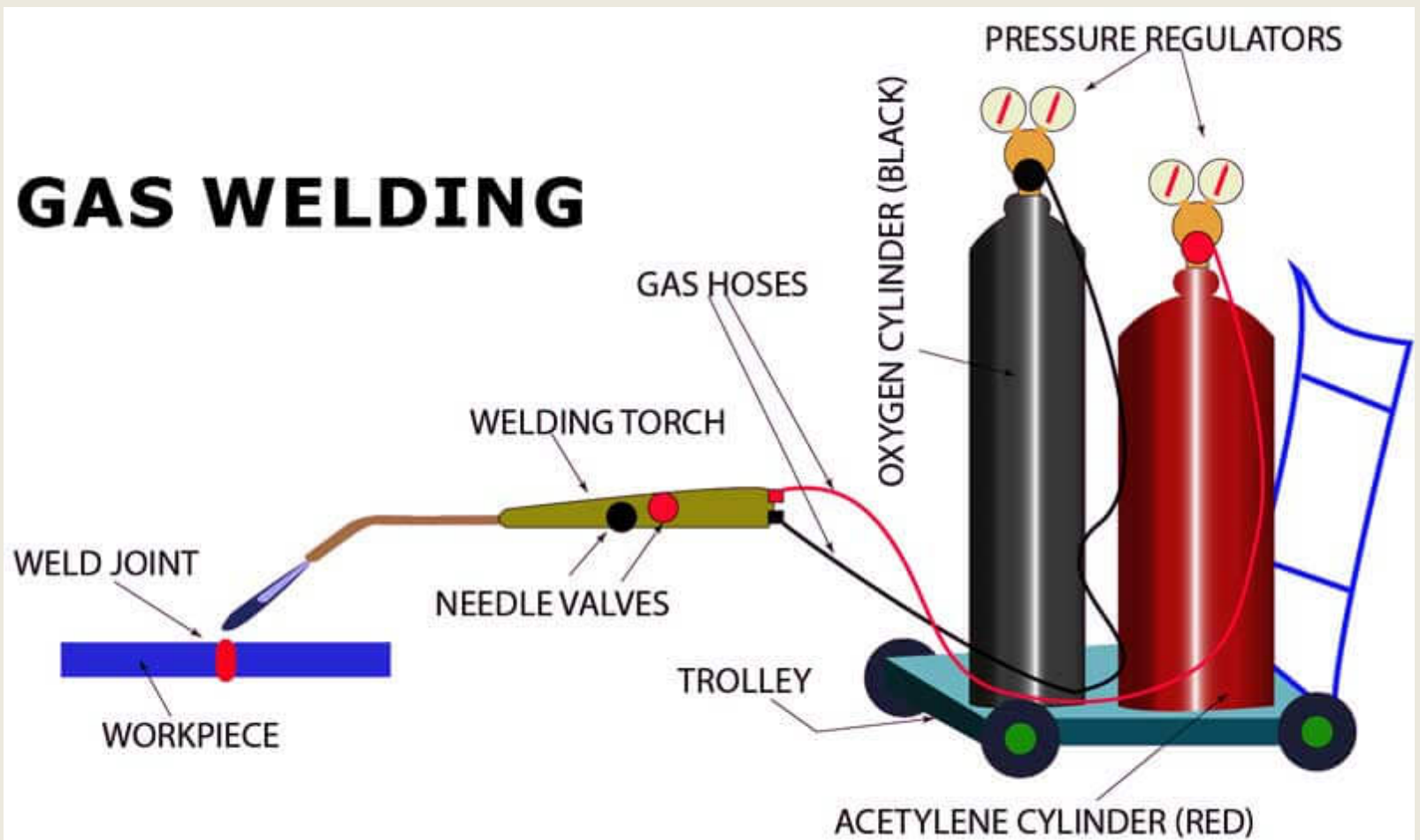
SPOT WELDER



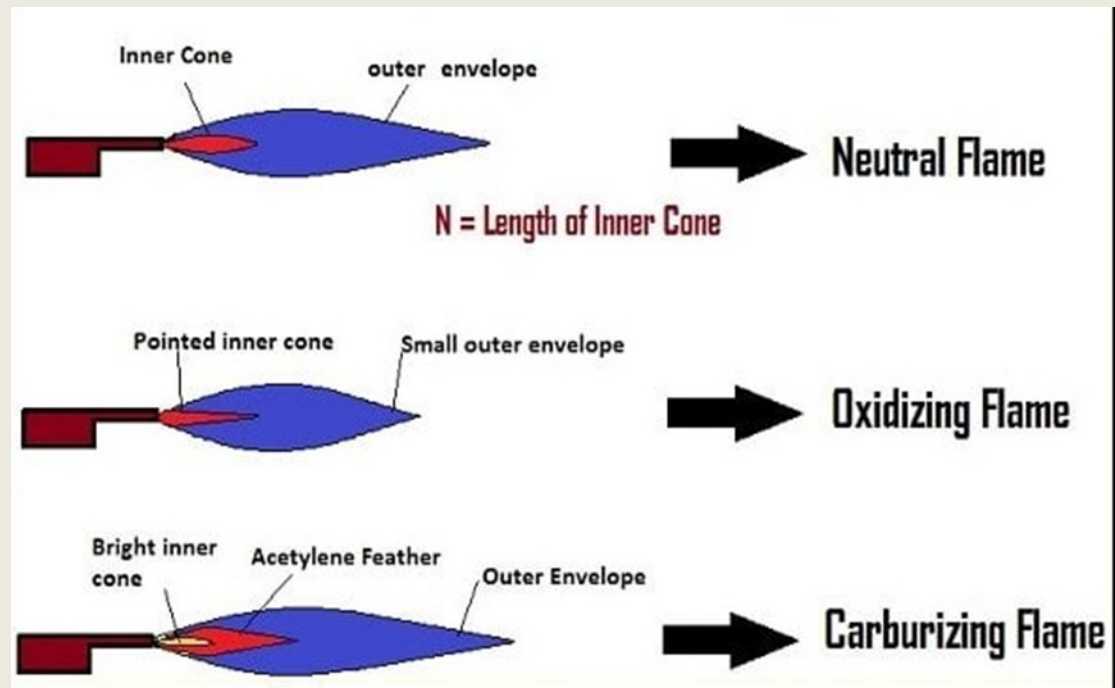
MIG Welder



Gas Welding



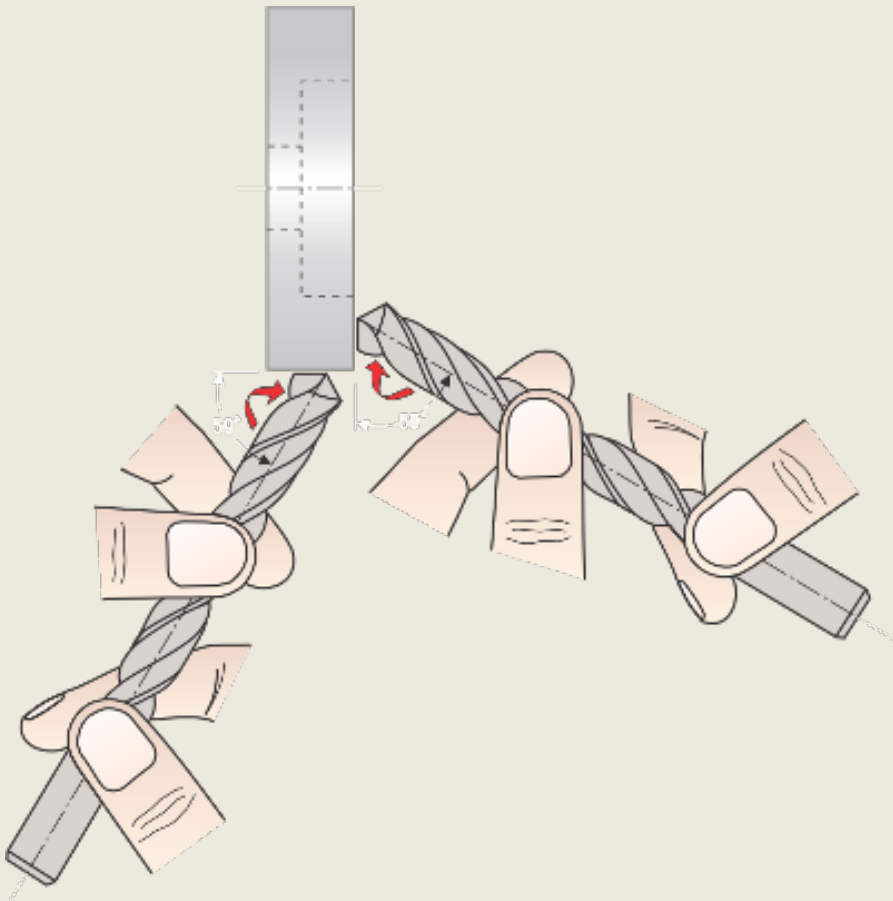
Gas Welding



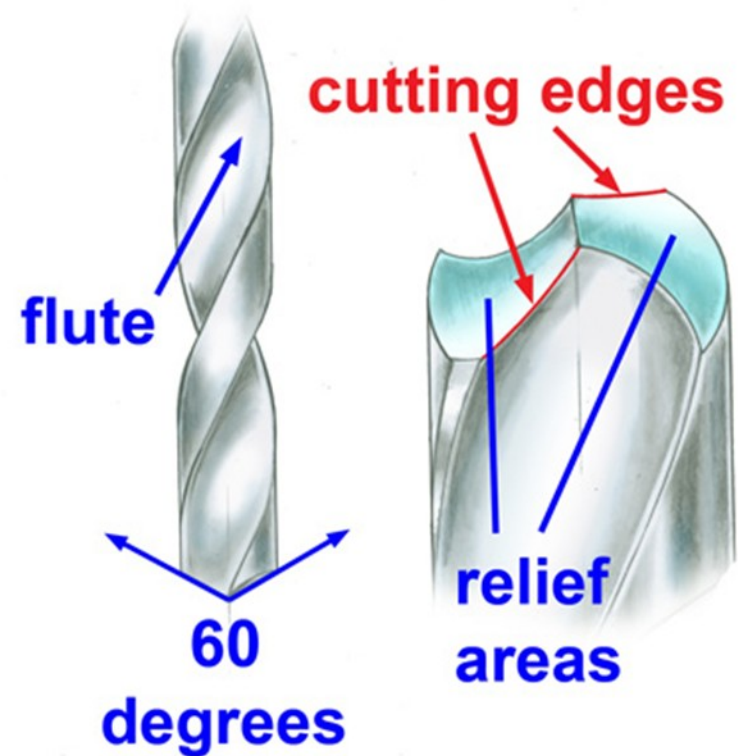
POWER DRILL



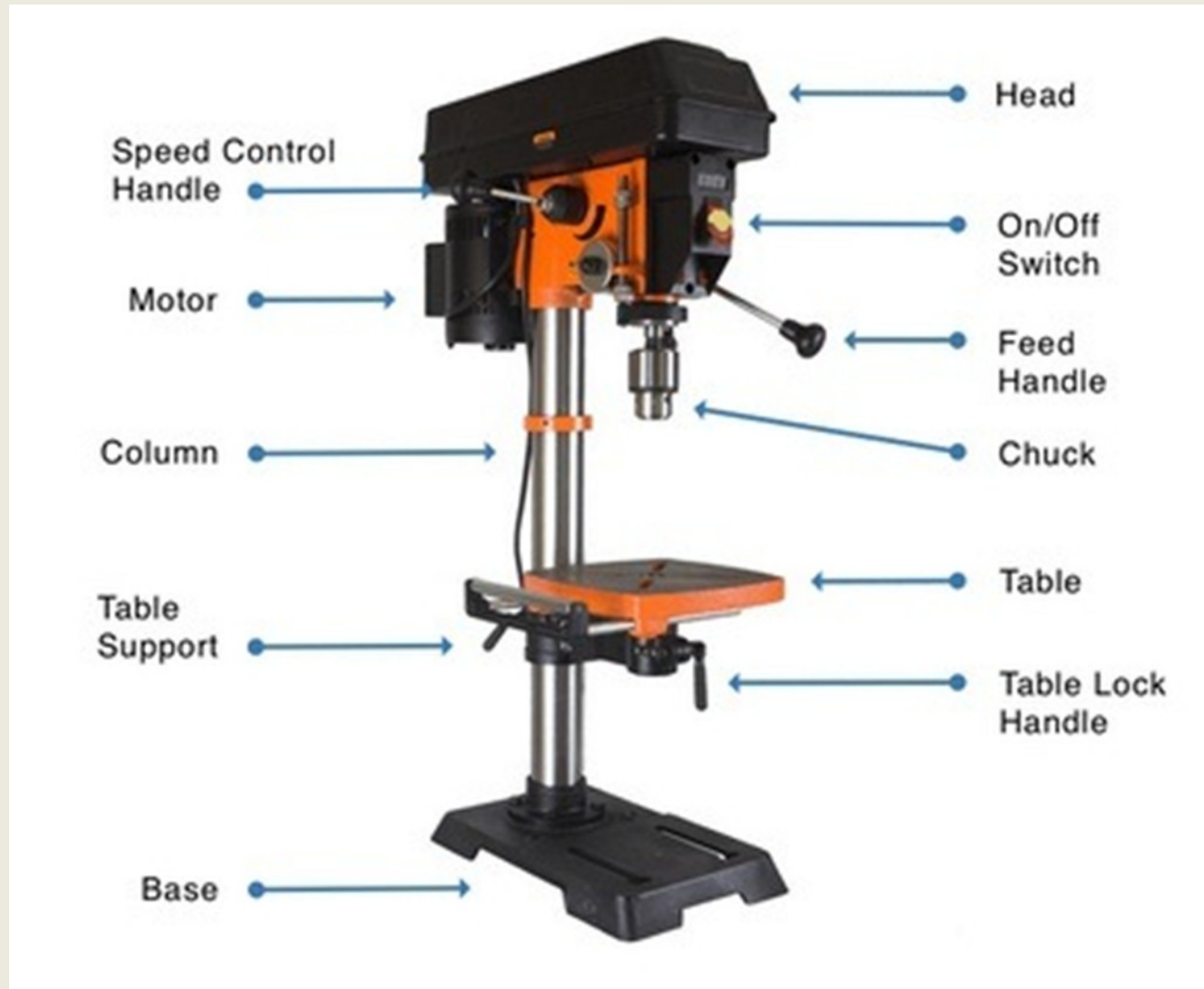
Sharpening Drill Bits



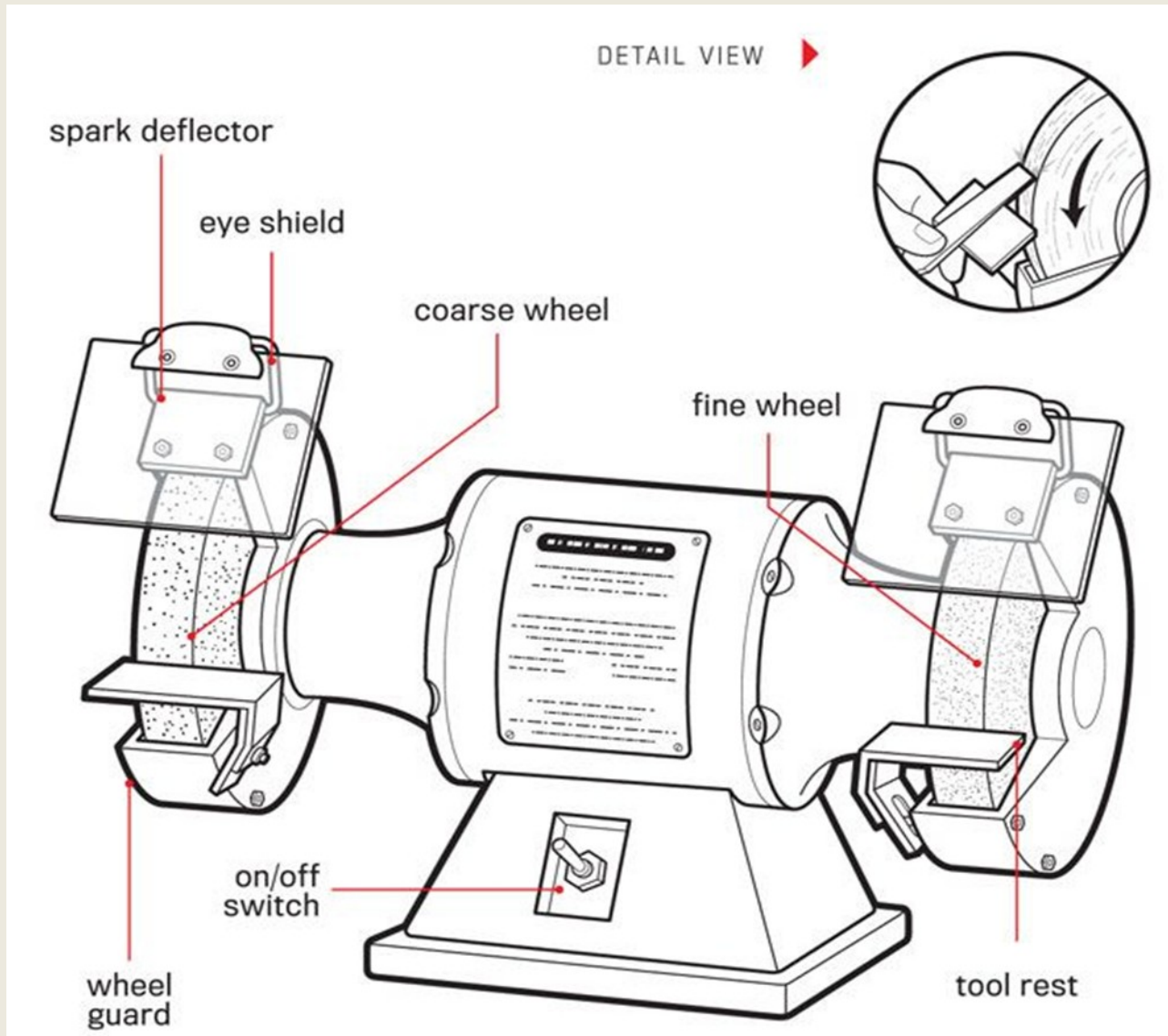
Parts of a Drill Bit



The Drill Stand

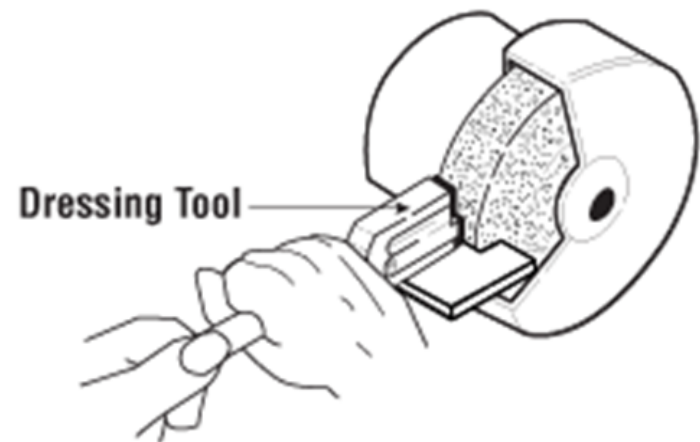
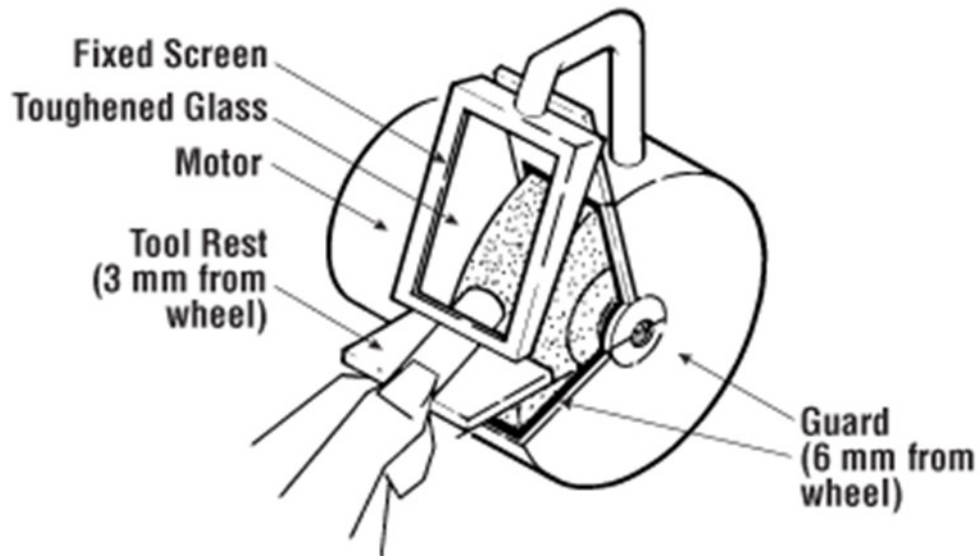


Bench Grinder





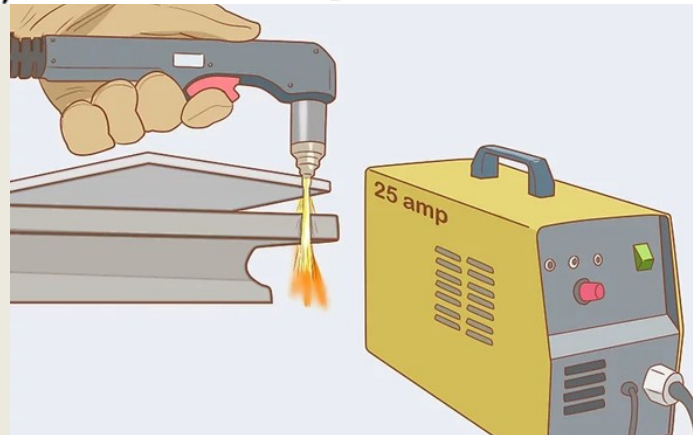
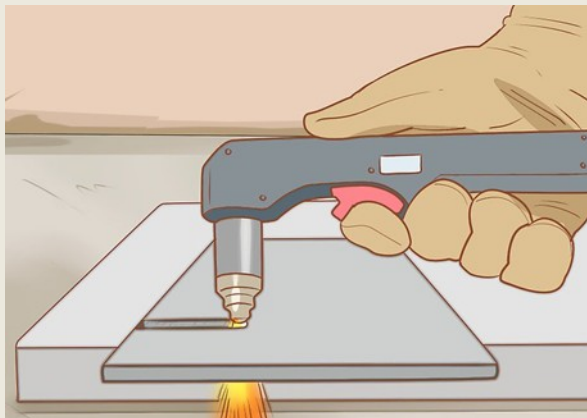
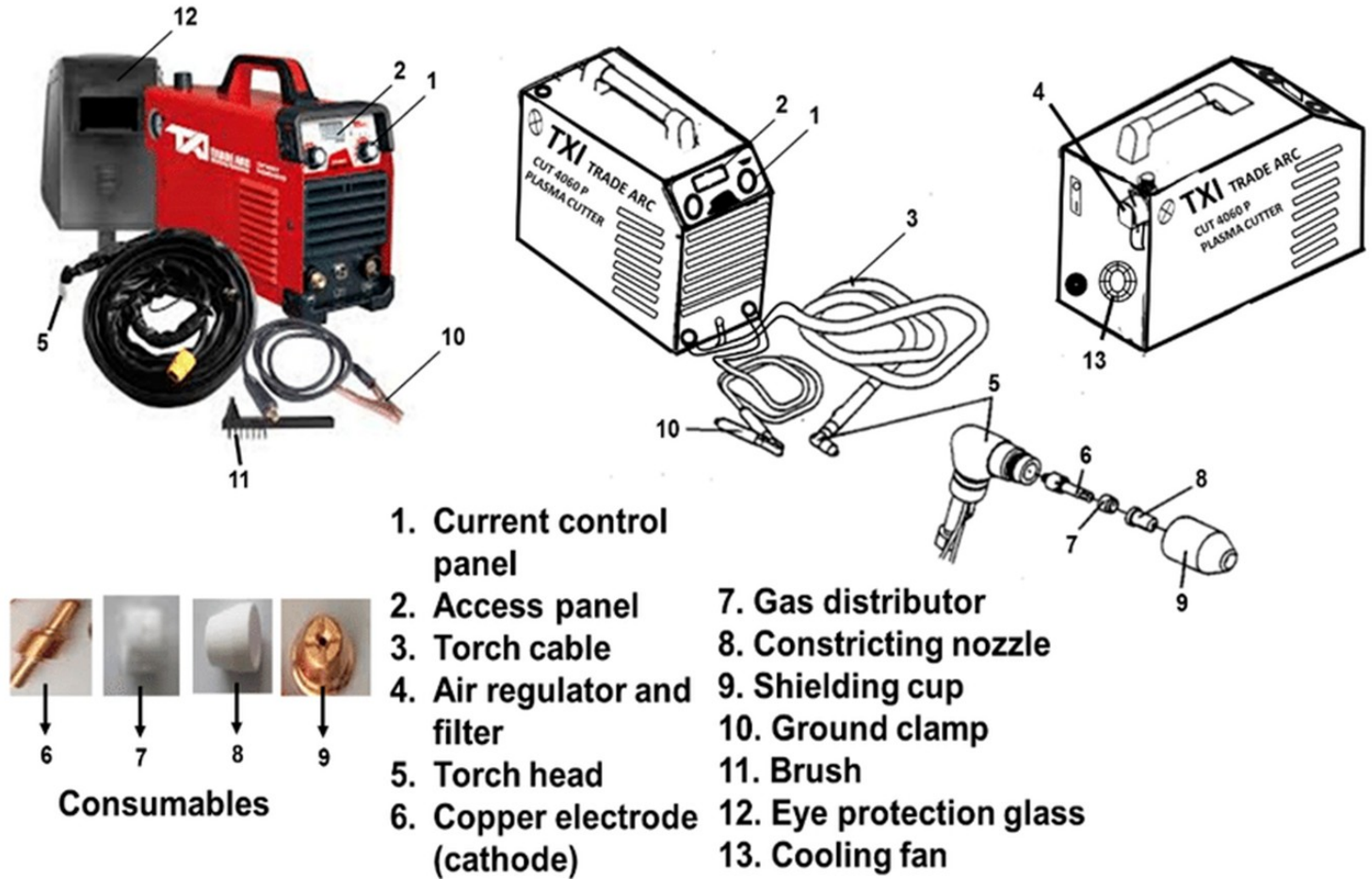
Bench Grinder



Angle Grinder



Plasma Cutter



Shearing Machine

